

# **Data visualization** guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 0.2 (pilot) September 2026 All example data is synthetic

OVERVIEW

# How to use this guide

One set of rules, three ways in. The guide teaches the thinking; the starter kit makes the right answer the easy one in every tool we use.

Every chart we make should do one job: answer a business question so clearly that the reader knows what to do next. The rules here are written to be checked, by a reviewer or by an agent, before a chart leaves the team.

Inspired by the structure of a leading consulting firm's public data-visualization guide, rebuilt for McKesson's brand and for the work of Workforce Innovation.

### CONSULTANTS AND ANALYSTS

**Decks, reports, workbooks**

Read Fundamentals once, then use the chart pages as you build. Keep the cheat sheet next to you.

- PowerPoint template and starter deck
- Excel chart workbook with the palette
- One-page cheat sheet and chart chooser
- Power BI report theme

### DEVELOPERS AND DATA SCIENTISTS

**Python, HTML, R, Power BI**

Each chart page links to tested code. Themes are generated from one token file, so every tool matches.

- `mckviz` for matplotlib and Plotly
- CSS variables, Plotly and Vega-Lite themes
- `theme_mck()` for ggplot2
- `mv.lint(fig)` to check a chart

### CODING AGENTS

**Claude, Copilot and others**

Agents read `AGENTS.md` automatically in this repo. For other repos, copy one self-contained rules file.

- `AGENTS.md`: workflow and rules
- `mck-chart` skill: question to QA
- `dist/agents/mck-viz-rules.md`
- Cookbook pages with runnable snippets

### The five steps behind every chart

**1 Question.** Write the question the chart answers in one sentence.

**2 Chart.** Pick the chart from the question type, not from habit.

**3 Build.** Grey first, then color only what the headline names.

**4 Headline.** Compute the finding, then write it as a sentence.

**5 QA.** Run the checklist and look at the rendered chart.

OVERVIEW

# Contents

Fundamentals apply to every chart. Each chart family then gets the same treatment: anatomy, variants, best practice and how to build it.

This is the pilot release. Sections marked NEXT follow the same pattern once the look and depth of the pilot are agreed.


01 **Fundamentals**
Question first, how readers read, anatomy, headlines, color, type, clutter, honest scales, people data, accessibility

02 **Magnitude**
Bar and column charts: anatomy, variants, best practice, build it

03 **Part-to-whole**
Stacked bars, waterfalls, treemaps, donuts: anatomy, variants, best practice, build it

04 **Change over time**
Line, stacked area, bubble timeline, small multiples

05 **Ranking**
Slope, bump, dot plot and dumbbell, heat table

06 **Distribution**
Histogram, strip, box plot

07 Correlation NEXT
Scatter and bubble, heatmaps, why not dual axes

08 Relationships and flows NEXT
Sankey, parallel coordinates, Venn

09 Geo-spatial NEXT
Choropleth, bubble map, dot density

10 Single number NEXT
KPI and big number, progress to target

11 Formats NEXT
Slides, reports and print, web and mobile, dashboards, interactivity, animation, social

12 Toolkit NEXT
Downloads, set-up for each tool, the QA checklist, guidance for agents

# **Start with the question,** not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

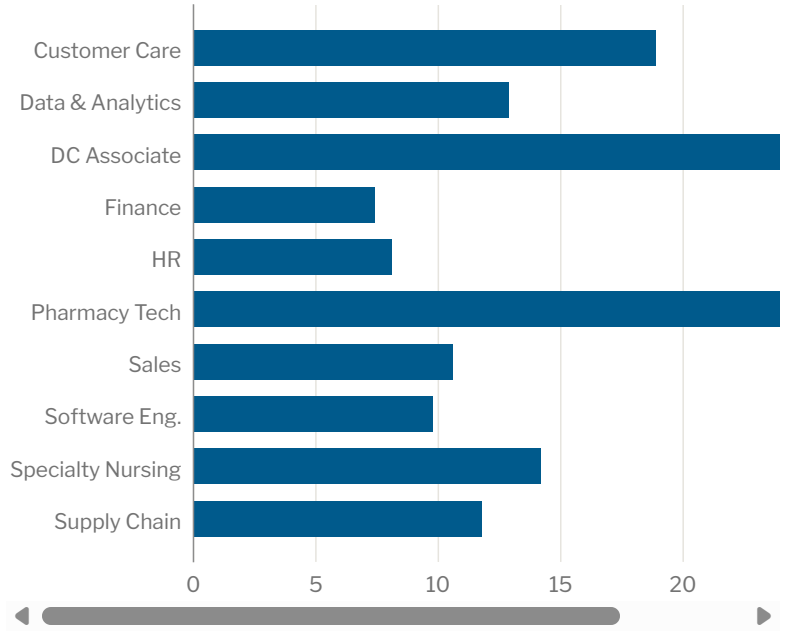
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

### Attrition by job family

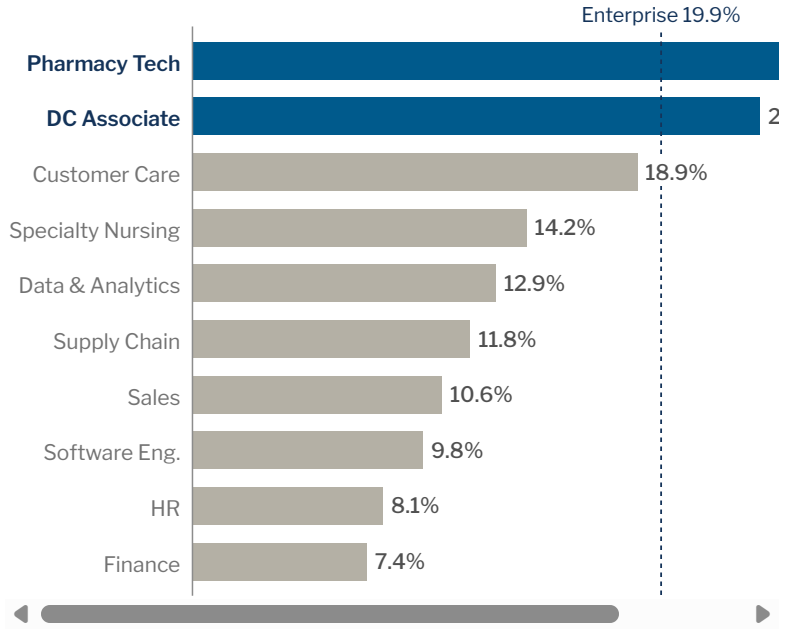
FY26




**Shown**
A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

### Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)




**Explained**
The headline states the finding, the order ranks the families, and color points to the two the headline names.

# **How readers** read a chart

Readers move through a chart in three steps. Design for each one, in order.

### 1 Perceive

In the first second the eye takes in shape, length and color before reading a word. Make the thing that matters the thing that stands out: one color against grey.

### 2 Interpret

Next the reader looks for meaning: headline, labels, units, benchmarks. The headline should confirm what they just saw, or sharpen it.

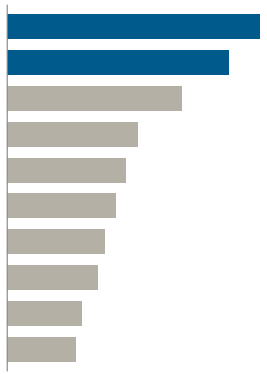
### 3 Act

Finally they ask "so what?". A callout that sizes the implication turns a finding into a decision.

STEP 1 · PERCEIVE

**Two bars stand out**

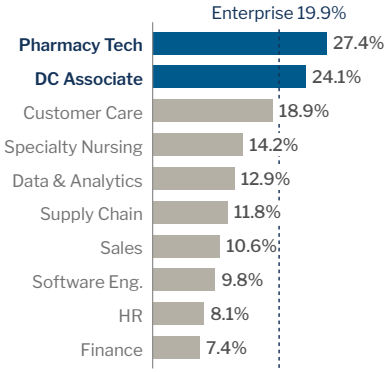
No words yet. Grey context and blue focus already tell the reader where to look.



STEP 2 · INTERPRET

**Labels give it meaning**

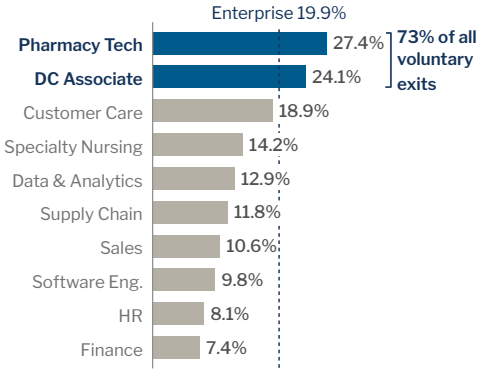
Names, values and a labelled benchmark show the two families sit well above 19.9%.




STEP 3 · ACT

**The callout sizes the prize**

Together they hold 58% of headcount but 73% of voluntary exits: a retention plan for two families moves the enterprise number.



# **Anatomy** of a chart

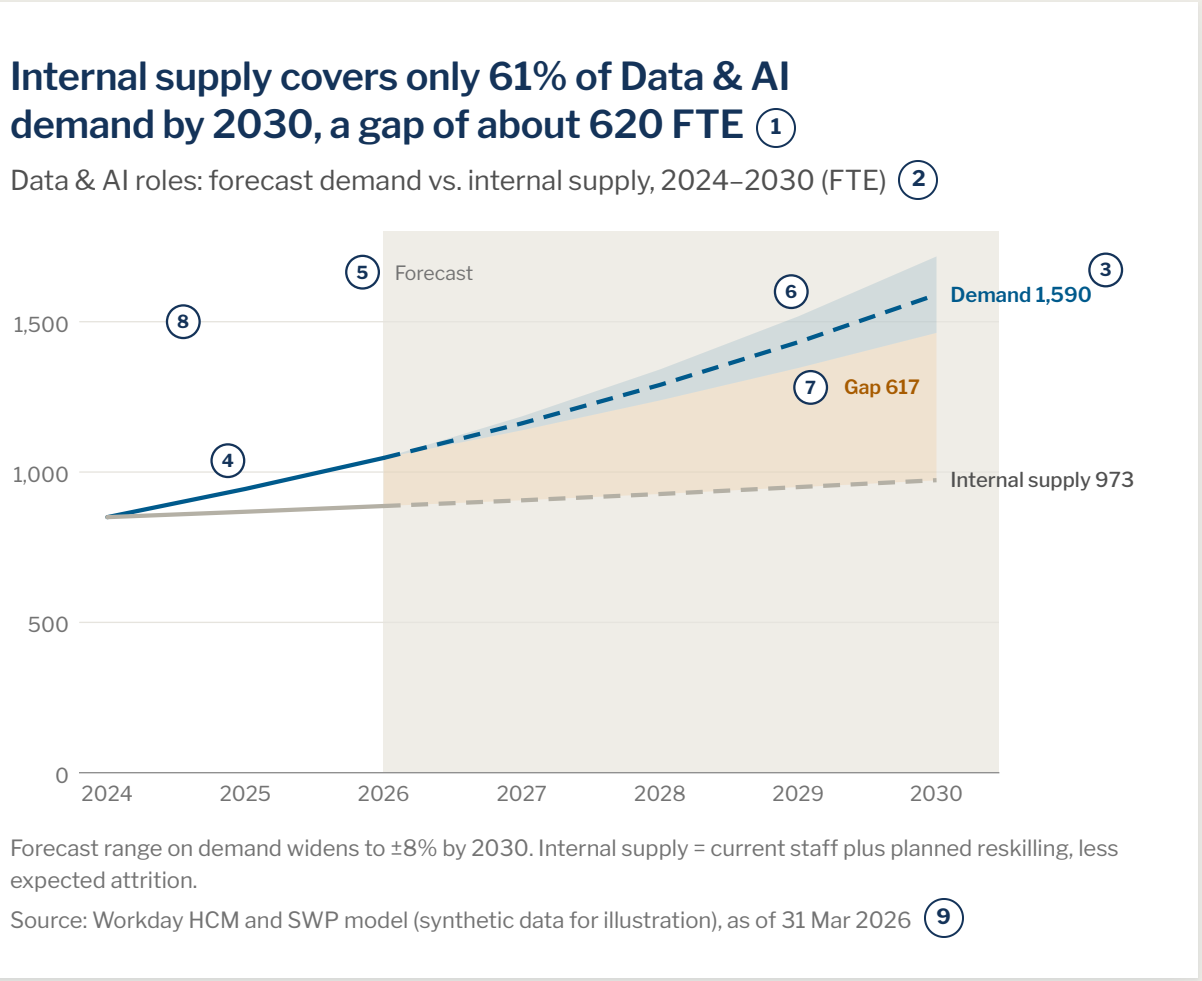
Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.



FUNDAMENTALS

# **Write the headline** last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

### The formula

**Who or what + verb + number + so what.**

### Before you keep it

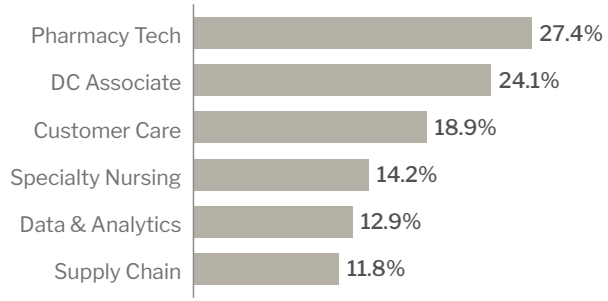
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

| | |
|---|---|
| **LABEL** | Attrition by job family · names the topic; says nothing |
| **VAGUE** | Attrition varies across job families · true of every chart ever made |
| **OPINION** | Attrition is out of control in the frontline · a claim the chart can't support |
| **INSIGHT** | **Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate** |
| **INSIGHT + ACTION** | **Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate** |

## Attrition varies across job families

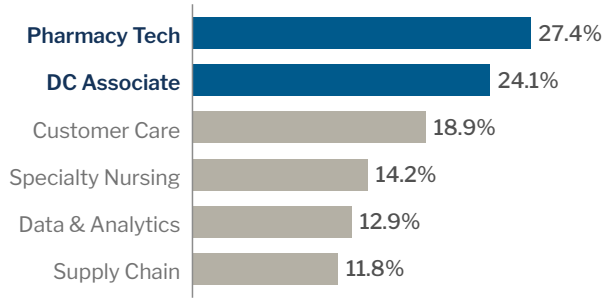
Voluntary attrition by job family, FY26 (%)




**Headline and chart both neutral**
Nothing tells the reader where to look.

## Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)




**Headline and highlight agree**
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

# **Grey first,** then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

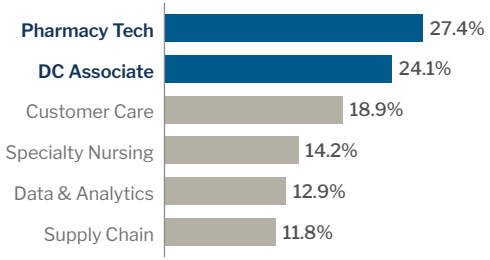
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

**Orange means "worse".** Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

### Focus

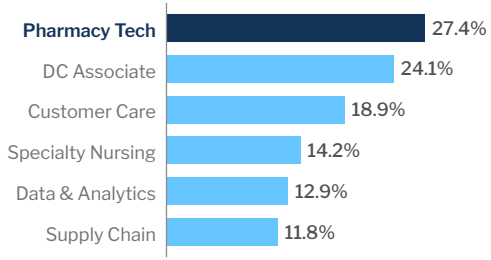
All grey; the items the headline names in blue.




PATTERN 2

### Uniform + accent

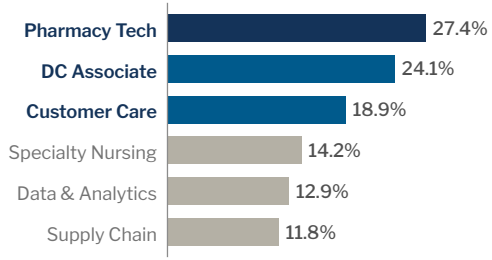
One series in light blue; the single exception in navy.




PATTERN 3

### Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.




### Color roles

| | | |
|---|---|---|
| **Context** · everything the reader should see but not focus on | `#B4B0A5` |
| **Focus** · what the headline names | `#005A8C` |
| **Accent** · the one exception that must pop (navy, not orange) | `#143359` |
| **Light blue** · the series when navy marks the exception | `#66C4FF` |
| **Orange** · only when it means worse: below target, shrinking, a shortfall | `#EF8200` |
| **Total and benchmark** · totals, targets, reference lines | `#143359` |
| **Context light** · de-emphasized fills in stacks and areas | `#DEDBD1` |

# **Three palettes,** three jobs

Pick the palette from what the color encodes.

### Categories

Only when categories must be told apart. At most six, always in the same order, so "Full-time" is the same blue on every slide.

### One measure, light to dark

For heat tables and choropleths of a single measure. Darker means more.

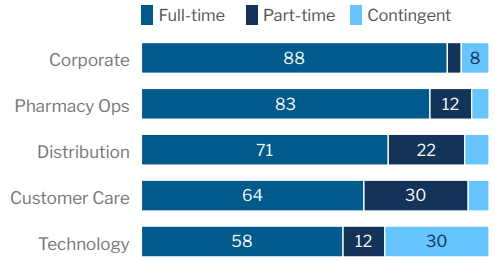
### Gap to a benchmark

For values centered on zero or a target. Orange below, blue above, a neutral middle.

Text on a fill switches between white and navy, whichever contrasts more (`mv.text_on`).

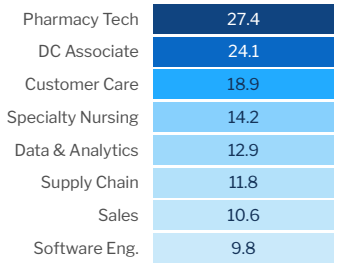
**CATEGORICAL**

Workforce mix by function (%)



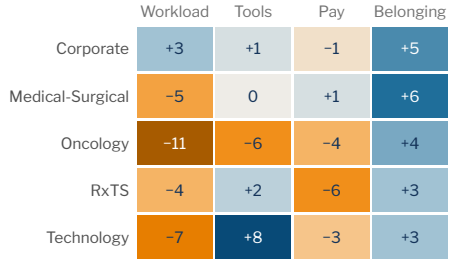

**SEQUENTIAL**

Voluntary attrition, FY26 (%)




**DIVERGING**

Engagement vs. healthcare benchmark (pts)




**Categorical**
Max 6, always this order

005A8C 143359 66C4FF 00AF9D 00737A EF8200

**Sequential**
One measure, light to dark

E6F3FB B6E2FA 66C4FF 009EFF 0A5FBC 143359

**Diverging**
Centered on 0 or a target

A75B00 EF8200 F7C180 EFEDE7 99BDD1 005A8C 143359

# **Type** and proportion

One typeface, Libre Franklin, in a few fixed roles. Type should frame the data, not compete with it.

The headline is the largest text on the chart but the data is the largest *thing*. If the headline is the first thing you notice from across the room, it is too heavy.

- Nothing below 9 pt on slides, 11 px on screens
- Sentence case everywhere; no all-caps headlines
- Headline, subtitle, key, footnote and source share one left edge
- Big numbers use the light weight

Libre Franklin is open source (SIL OFL). Set-up for Office, the web and Power BI is in the Toolkit.

| Role | Sample | Weight | Slide | Screen |
|---|---|---|---|---|
| Headline | **Attrition is concentrated in two job families** | Bold | 24 pt | 22 px |
| Subtitle | Voluntary attrition, FY26 (%) | Regular | 14 pt | 15 px |
| Data labels | 27.4% | Medium | 12 pt | 13 px |
| Axis and category labels | Pharmacy Technician | Regular | 11 pt | 12 px |
| Callouts | Hiring freeze, Q2 | Regular | 12 pt | 13 px |
| Footnote and source | Source: Workday HCM, as of 31 Mar 2026 | Light | 9 pt | 11 px |
| Big number | 75% | Light | 54 pt | 56 px |

### Proportion on a 16:9 slide

Margins of about 0.5 in. The chart takes at least 60% of the slide height. One message per chart, at most two charts per slide; the slide title is the headline.

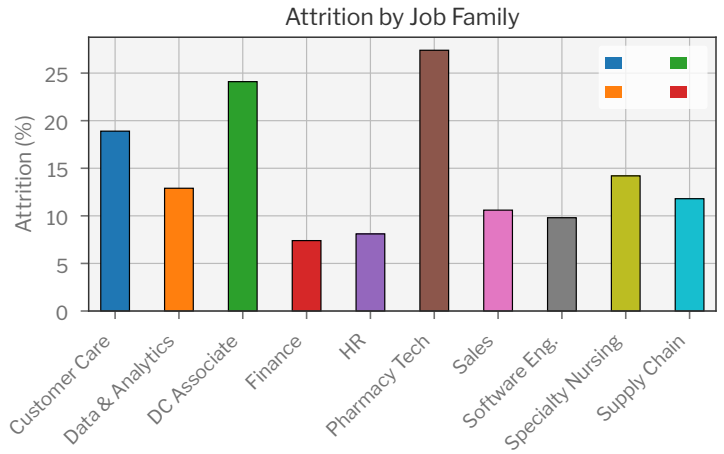
### Proportion on a screen

The chart card sits on a quiet background with generous padding. Headline about 1.4× the subtitle. On phones, swap end labels for a top key and let wide exhibits scroll sideways rather than shrink.

# **Label directly,** delete the rest

Every element must earn its place. If removing it loses nothing, remove it.

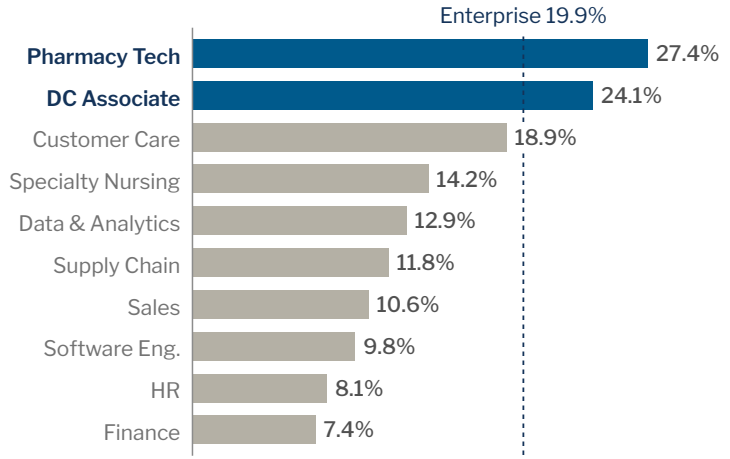
1. Replace the label-style title with an insight headline
2. Sort by value, not alphabet
3. Turn long-named columns into horizontal bars
4. Grey everything, then color the point
5. Label values in place; delete the value axis, gridlines and tick marks
6. Delete the legend, borders and background fill
7. Add the benchmark, labelled where it sits




**Tool defaults**
Rainbow fills, borders, rotated labels, gridlines, a legend that decodes nothing.

**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate**

Voluntary attrition by job family, FY26 (%)




**Decluttered**
The same data. Fewer elements, and every one of them carries information.

FUNDAMENTALS

# **Honest** scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

**Bars start at zero.** If the differences only show on a narrow range, change the chart, not the axis.

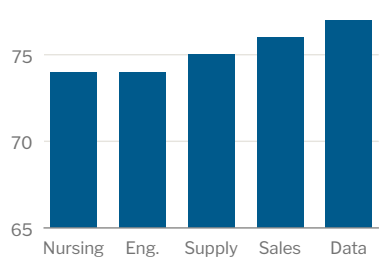
**No dual axes.** Two scales let you make any two lines cross wherever you like.

**Bubble area, not diameter,** is proportional to value.

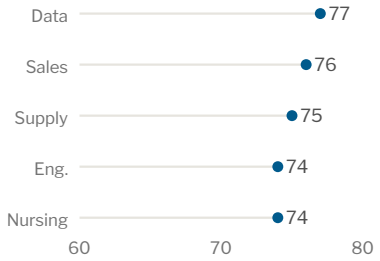
**Pies take five slices at most,** and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

## Truncated bars

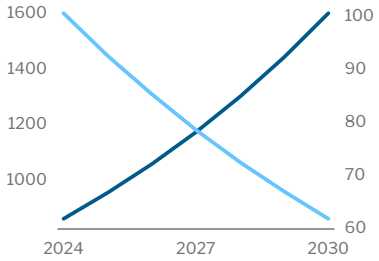



**Try to avoid**
An axis from 65 turns a 4% difference into a 33% one.

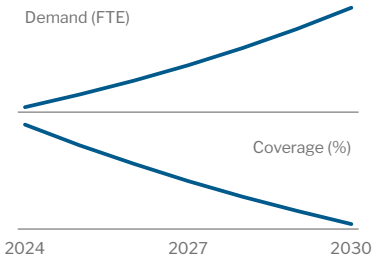



**Recommended**
A dot plot compares position, so the range can be narrow.

## Dual axes

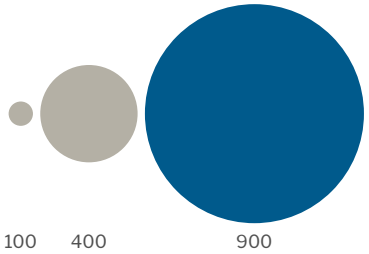



**Try to avoid**
The crossing point is an artifact of two chosen scales.

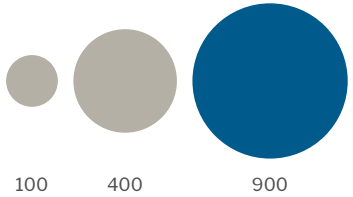



**Recommended**
Two aligned charts share the time axis.

## Bubble size

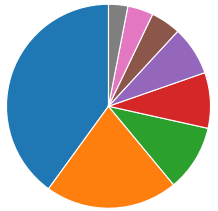



**Try to avoid**
Sized by diameter, 900 looks 81× bigger than 100.

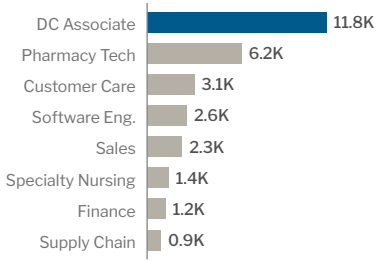



**Recommended**
Sized by area, 900 is 9× bigger, as it should be.

## Too many slices



**Try to avoid**
Eight similar angles can't be ranked by eye.




**Recommended**
A sorted bar ranks them instantly.

# **People data** is personal

Workforce charts describe colleagues, and carry extra obligations.

### Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

### No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

### Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

**Engagement by team, one distribution center**

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

| Team | Respondents | Engagement |
|---|---|---|
| Inbound | 46 | 71 |
| Outbound | 38 | 68 |
| Returns | 7 | 52 |
| Quality | 4 | 88 |
| Site leadership | 9 | 81 |

**Try to avoid**
Reporting groups below the minimum, even in a table or tooltip.

**Engagement by team, one distribution center**

Small teams combined into one group that clears the threshold.

| Team | Respondents | Engagement |
|---|---|---|
| Inbound | 46 | 71 |
| Outbound | 38 | 68 |
| Other teams (3 combined) | 20 | 72 |

**Recommended**
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

### In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

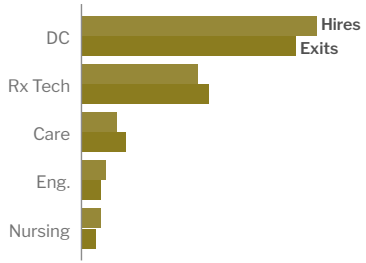
# **Accessible** by default

About 1 in 12 men and 1 in 200 women have a color-vision deficiency. In a room of 25, assume someone does.

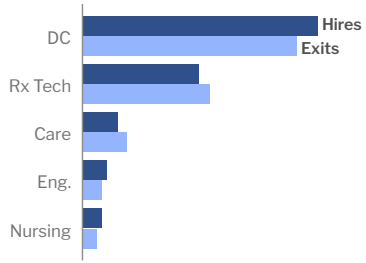
- Text contrast at least 4.5:1
- Never red vs. green; use blues and navy, with orange only for "worse"
- Never color alone: pair it with labels, position or shape
- Nothing below 9 pt on slides or 11 px on screens
- Alt text for every chart that leaves the team: the headline plus two or three key numbers

The red–green examples are simulated for deuteranopia, the most common form.

## Red vs. green

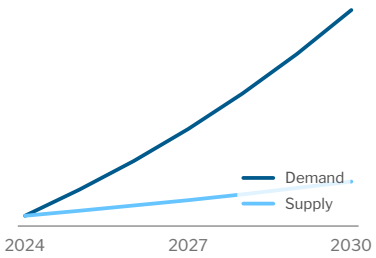



**As seen with deuteranopia**
Red and green hires vs. exits become one color.

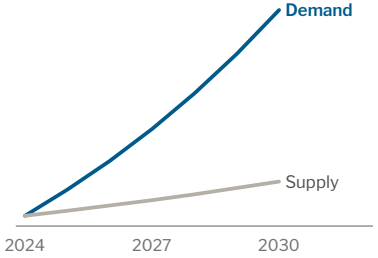



**As seen with deuteranopia**
Blue and light blue stay distinct.

## Color as the only cue

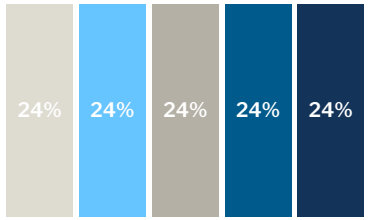



**Try to avoid**
The reader matches colors to a legend.

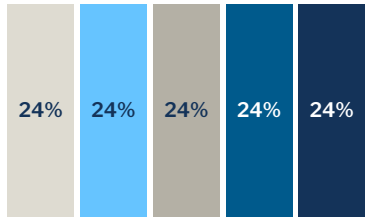



**Recommended**
Names at the line ends; color only reinforces.

## Text on fills




**Try to avoid**
White on every fill fails on the light ones.




**Recommended**
White or navy, whichever contrasts more.

## Alt text

Write it from the headline and the numbers that prove it.

> **Alt:** Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate. Pharmacy Technician 27.4%; Distribution Center Associate 24.1%; all other job families are below the enterprise rate.

In PowerPoint: right-click the chart, View Alt Text. In HTML: `aria-label` on the figure.

MAGNITUDE

# **Bar** chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

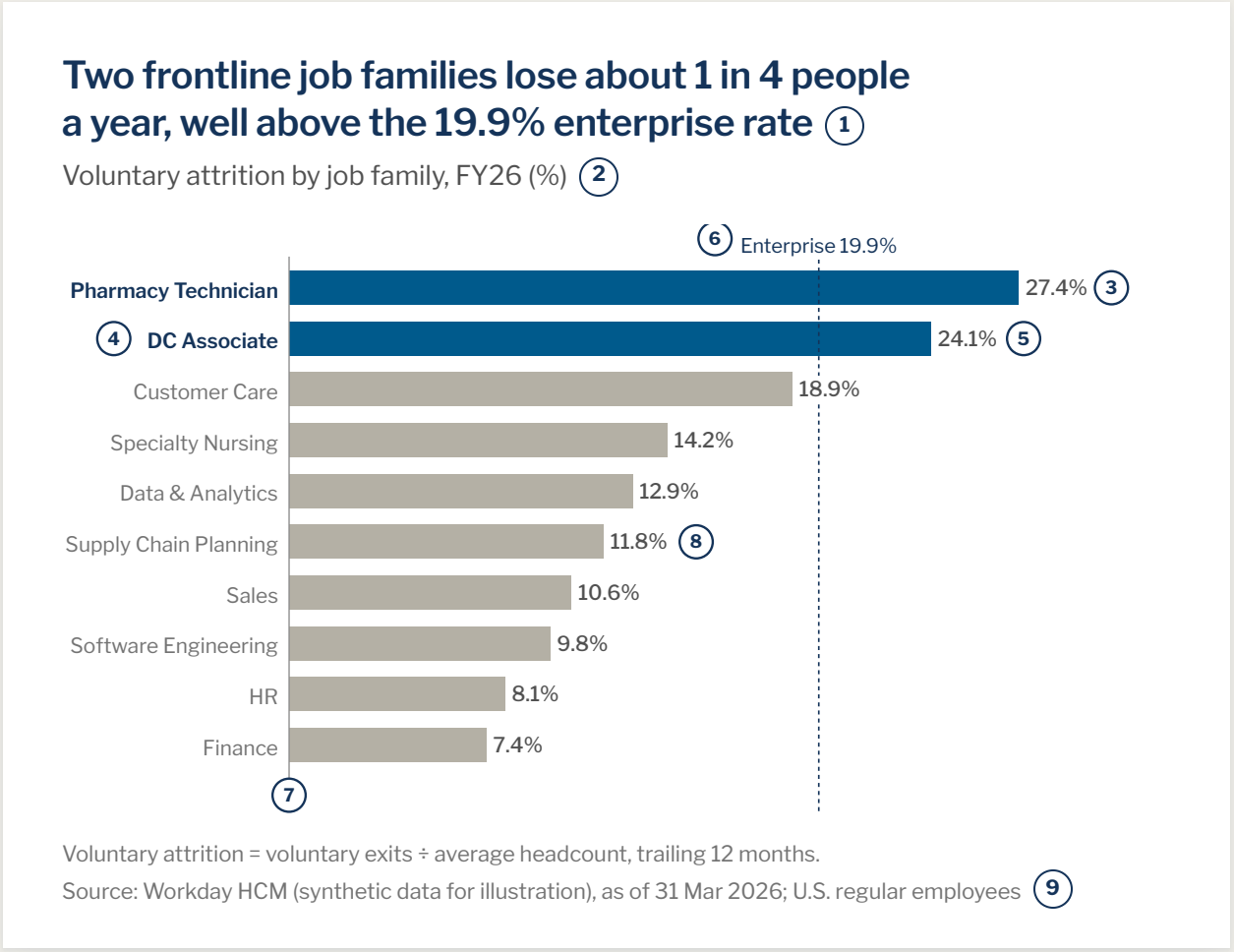
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

### McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

# Bar chart variants

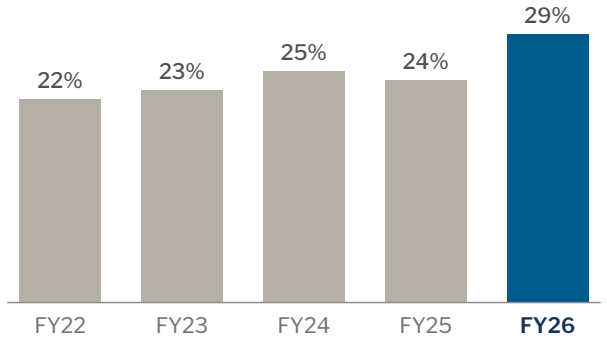
Same rules, different shapes. Pick the variant from the comparison you need to make.

### When a bar is the wrong chart

- Values bunched far from zero (engagement 71–78): use a **dot plot**
- Largest more than 20× the smallest: use **proportional bubbles** or split the chart
- Parts of a whole: use a **100% stacked bar** or **waterfall**
- Many periods: use a **line**
- Spread across people: use a **distribution**

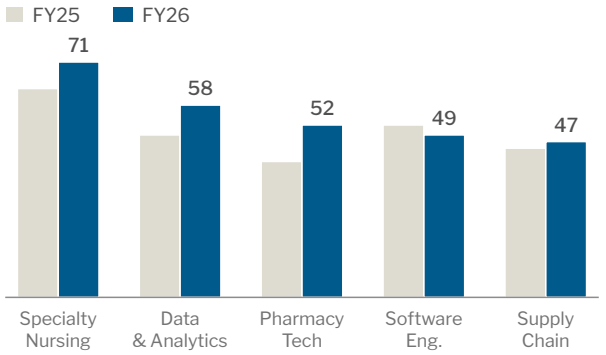
### Column chart

A few periods in time order, or short category names. Here internal fill rose from 22% to 29%.



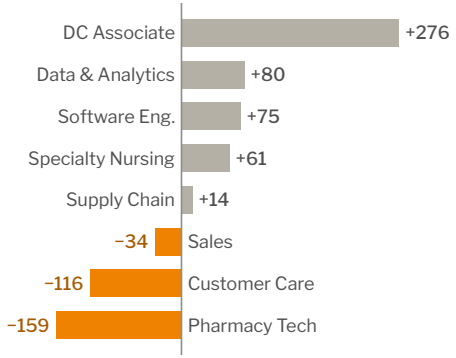

### Grouped bars

Two or three series per category. Prior period in light grey, current in blue; key top-left.



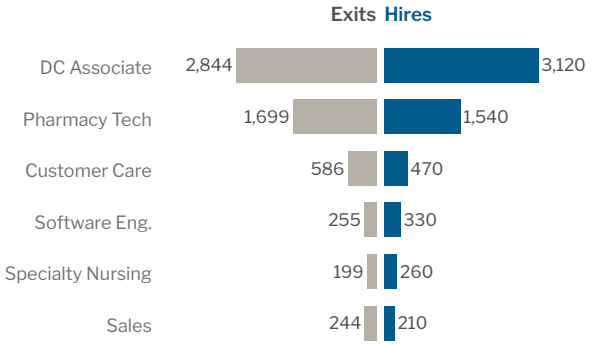

### Diverging bars

Values on both sides of zero. Three job families shrank in FY26 despite hiring; negatives in orange.




### Butterfly chart

Two measures back to back on the same items: FY26 exits and hires by job family.



MAGNITUDE

# **Bar chart** best practice

Four rules that fix most bar charts.

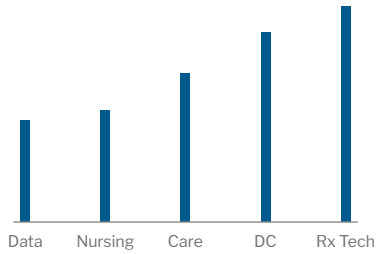
**Size of bars.** Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

**Color of bars.** A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

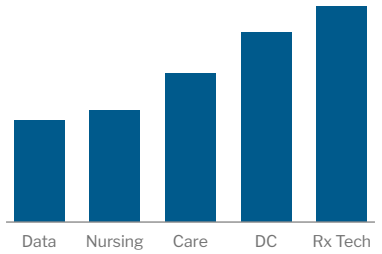
**Direct labels.** Gridlines and a value axis add noise. Label the bars and delete both.

**Zero baseline.** Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

## Size of bars

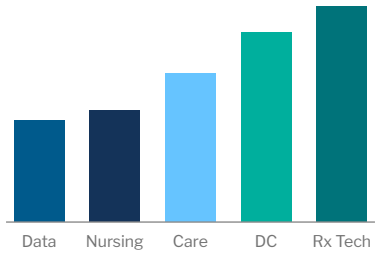



**Try to avoid**
Stick-thin bars; the gaps dominate.

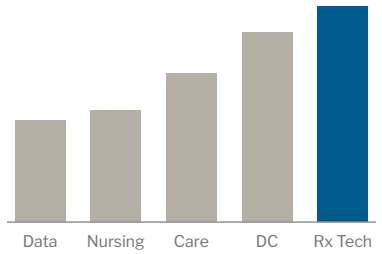



**Recommended**
Bar about twice the gap.

## Color of bars

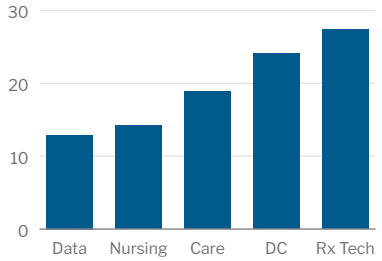



**Try to avoid**
A color per bar, encoding nothing.

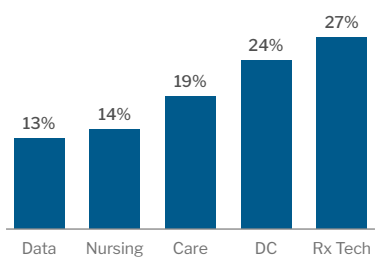



**Recommended**
Grey bars; blue for the one that matters.

## Direct labels

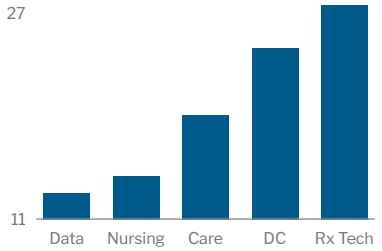



**Try to avoid**
Gridlines and an axis to read values from.

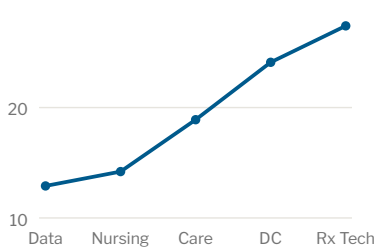



**Recommended**
Values on the bars; no axis, no grid.

## Zero baseline




**Try to avoid**
A truncated axis exaggerates differences.




**Recommended**
A line or dot plot can use a narrow range.

# **Bar chart** best practice, continued

Four more, from reviewing workforce decks.

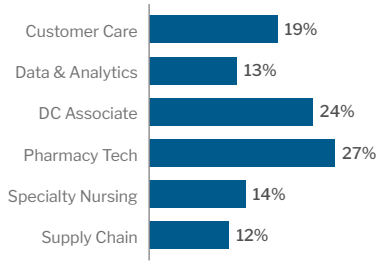
**Sort with intent.** Alphabetical order hides the ranking. Sort by value unless the categories have a natural order: tenure bands, job levels, survey scales.

**Horizontal for long names.** Rotated labels are slow to read. Job-family names need horizontal bars.

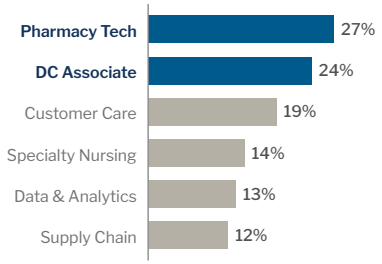
**Label the benchmark in place.** A reference line explained in a legend makes the reader search. Name it on the line.

**Two highlights at most.** Five colored bars in two colors is decoration, not emphasis.

## Sort with intent

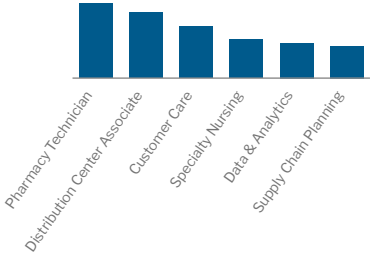



**Try to avoid**
Alphabetical: the ranking is hidden.

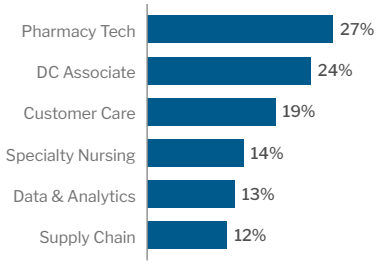



**Recommended**
By value, largest on top.

## Horizontal for long names

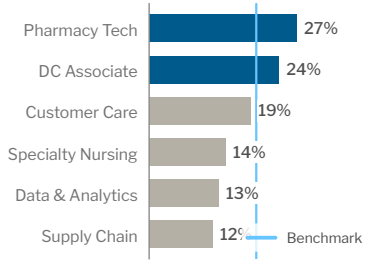



**Try to avoid**
Tilted names the reader has to decode.

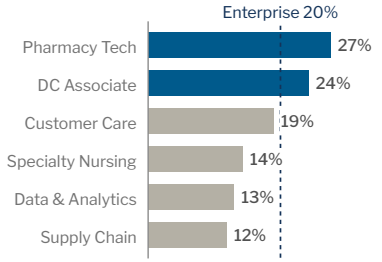



**Recommended**
Horizontal bars; names read naturally.

## Label the benchmark

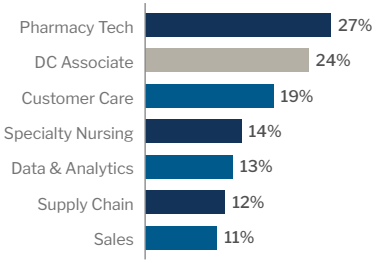



**Try to avoid**
A legend for one line, in a highlight color.

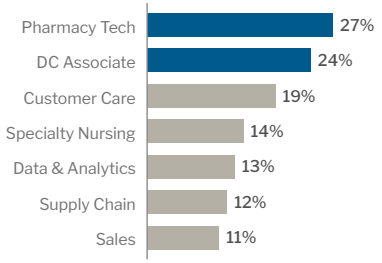



**Recommended**
Navy dashed line, named on the line.

## Two highlights at most




**Try to avoid**
Everything highlighted means nothing is.




**Recommended**
The two the headline names.

# **Build** a bar chart

The same chart in every tool we use. Each recipe starts from the starter kit, so the colors and fonts are already right.

Full, tested code for each platform is on the cookbook page: docs/cookbook/bar.md.

### Before you send it

- [ ] Starts at zero; bar about 2× the gap
- [ ] Sorted by value or a natural order
- [ ] Values labelled; no value axis or gridlines
- [ ] At most two highlights, matching the headline
- [ ] Benchmark labelled in place
- [ ] No group under 10 people

| | |
|---|---|
| **PowerPoint and Excel** | Start from the starter deck or chart workbook in `dist/office/`. Insert a clustered bar, sort the data ascending (so the largest is on top), set **Gap Width 50%**, fill all bars with Context grey and the focus bars with Focus blue. Add data labels at Outside End; delete the value axis, gridlines, legend and chart title. The slide title is the headline. |
| **Power BI** | Import `mckesson-dataviz-theme.json` and sort the bars by the measure. Drive the highlight from a measure, not by clicking bars: a `Bar color` measure returns the focus blue when attrition is above 20% and context grey otherwise, applied through **Colors › fx › Field value**. Turn on data labels; turn off the X axis and gridlines. See docs/powerbi-setup.md. |
| **Python** | matplotlib with `mckviz`; Plotly uses the same helpers (`mv.pl_highlight`). |

```
bars = ax.barh(df.job_family, df.rate, height=0.67,
               color=mv.highlight(df.job_family, focus=focus))
mv.label_bars(ax, bars, fmt="{:.1f}%")
mv.clean_axes(ax, keep=("left",), grid=None); ax.set_xticks([])
mv.titles(fig, headline, subtitle); mv.source(fig, source)
mv.lint(fig)   # checks colors, zero baseline, labels vs. axis
```

| | |
|---|---|
| **HTML** | Copy `templates/html/chart-card.html`: headline, subtitle and source are HTML, colors come from CSS variables such as `var(--mck-chart-focus)`, and the chart is drawn after the font loads. |
| **R** | `source("dist/r/theme_mck.R")`, then `geom_col(aes(fill = job_family), width = 0.67)` + `scale_fill_manual(values = mck_highlight(levels(df$job_family), focus = focus), guide = "none")` + `theme_mck()`. |
| **Coding agents** | "Make a McKesson bar chart of voluntary attrition by job family, FY26, highlighting the families above the enterprise rate." The `mck-chart` skill runs the workflow and returns the chart, a chart brief and alt text. |

PART-TO-WHOLE

# **Stacked** bar chart

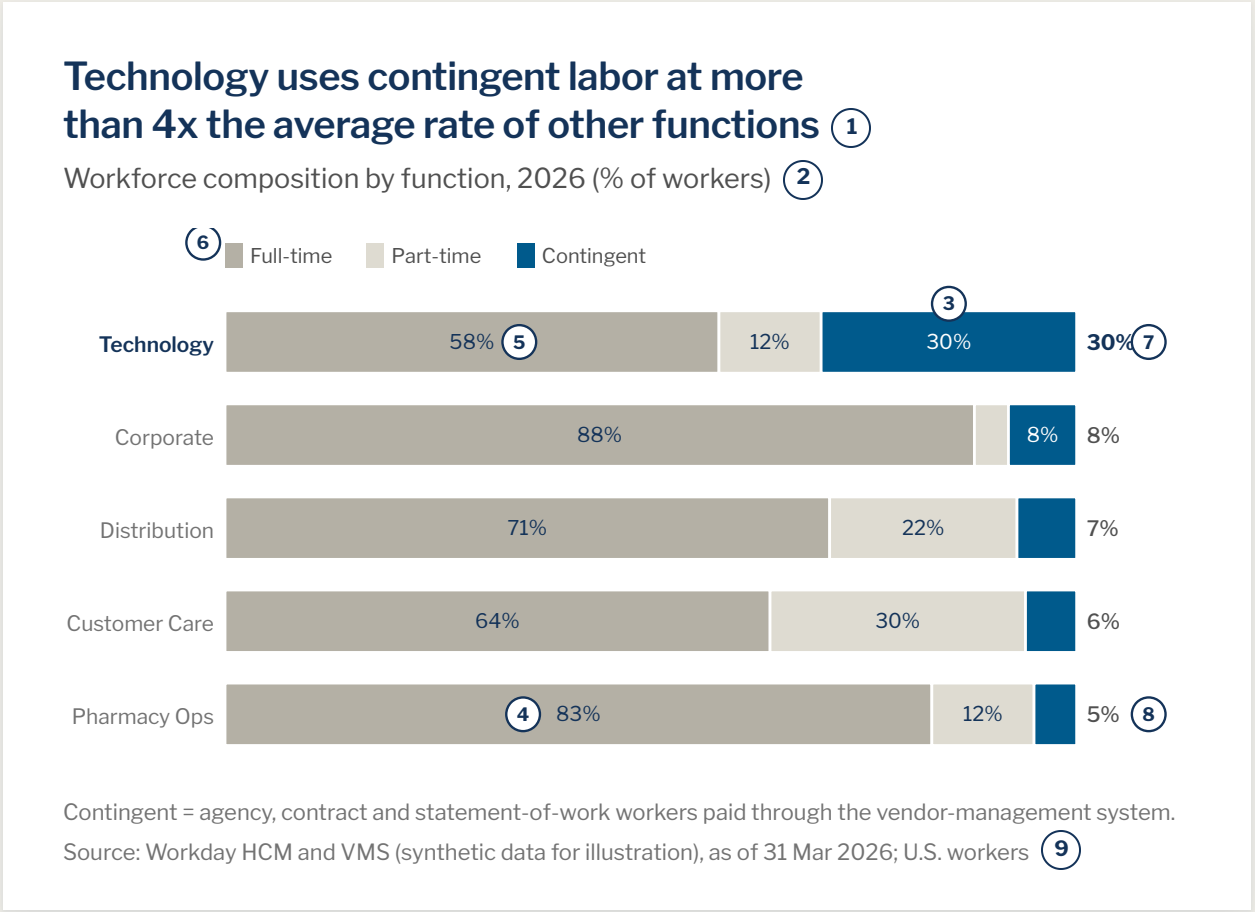
Part-to-whole charts show how a whole breaks into its parts, and how each part relates to the total.

A 100% stacked bar compares the mix of several wholes at once. Color only the segment the headline is about, and put it on the baseline or at the end of each bar.

### McKesson spec

- Five segments at most; group the rest as "Other"
- White separators between segments
- Labels inside segments that fit; key top-left in stack order
- Sort the bars by the segment that matters

1. **Insight headline**
   Names the part and how far it stands out.
2. **Subtitle**
   Metric, unit, scope, period.
3. **The part that matters**
   One segment in blue, at the end of every bar.
4. **Grey context**
   Two greys for the other parts.
5. **Labels in the segments**
   Where they fit; navy on grey, white on blue.
6. **Key in stack order**
   Top-left, left to right as the segments run.
7. **Echo the value**
   The story segment's value at the bar end, bold for the focus bar.
8. **Sorted by the story**
   Bars ordered by the blue segment, not alphabetically.
9. **Footnote and source**
   Define the parts.



PART-TO-WHOLE

# **Stacked bar** variants

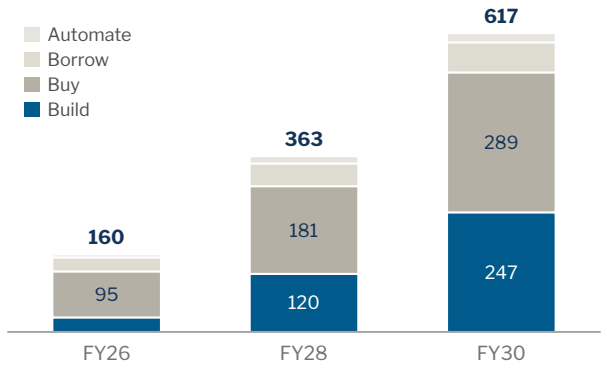
Same rules: one part in color, the rest grey, labels in place.

### When a stack is the wrong chart

- One whole with up to five parts: a **donut** or a single bar
- How a total changes step by step: a **waterfall**
- Many parts or a hierarchy: a **treemap**
- Comparing the parts themselves across groups: **grouped bars** or **small multiples**; only the baseline segment of a stack compares cleanly

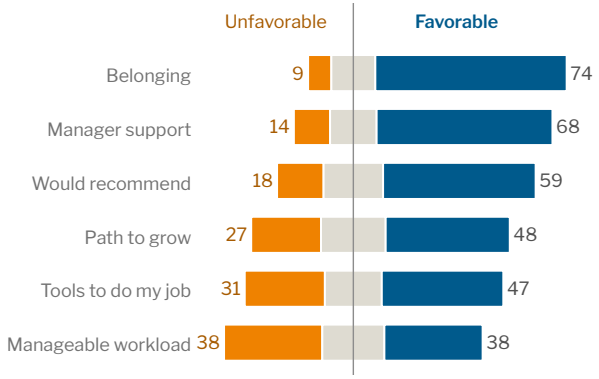
### Stacked columns with totals

When the total matters too. The FY30 Data & AI gap of 617 FTE: reskilling (on the baseline) closes 40%.



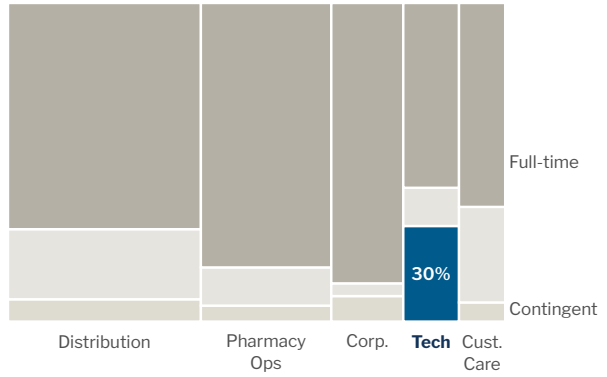

### Diverging stacked bars

Survey scales. Unfavorable in orange left of zero (it means worse), favorable in blue, neutral split across the middle.



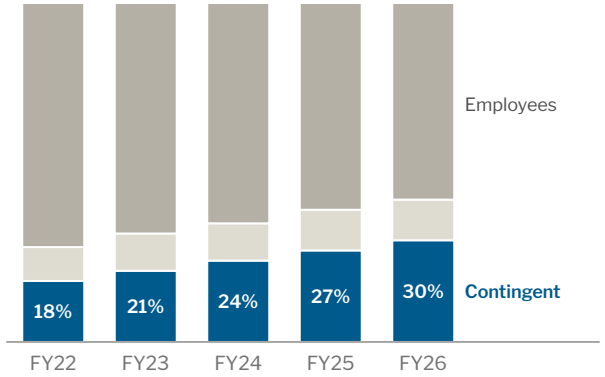

### Marimekko

Width and height both carry data: column width is each function's workforce, height its mix. Powerful, but it needs explaining.




### Mix over time

A few periods as 100% columns, the tracked segment on the baseline so its change reads directly.



PART-TO-WHOLE

# **Stacked bar** best practice

Stacks are easy to make and easy to misread. Four rules keep them honest.

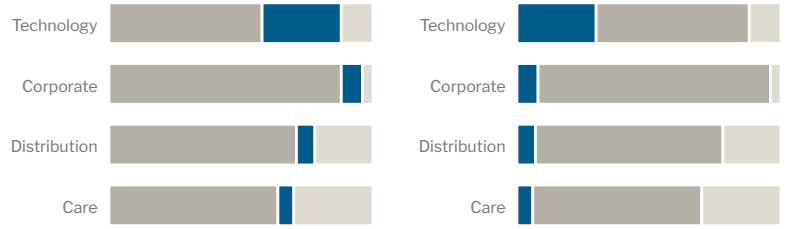
**The part that matters on the baseline.** Only segments that share a starting line can be compared by eye.

**Five segments at most.** More slices turn a bar into a barcode. Keep the parts the story needs and group the rest.

**Label in place; key in stack order.** If a key is needed, put it top-left, in the same order as the segments.

**One order in every bar.** Never re-sort segments within each bar.

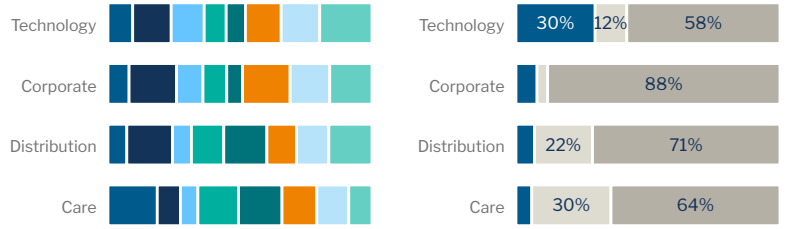
## The part that matters on the baseline




**Try to avoid**
A floating segment can't be compared across bars.

**Recommended**
On the baseline, every bar starts level.

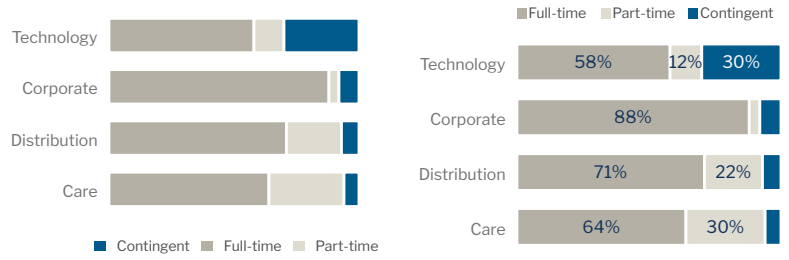
## Five segments at most




**Try to avoid**
Eight colored slices nobody can follow.

**Recommended**
The part in blue, the rest in grey.

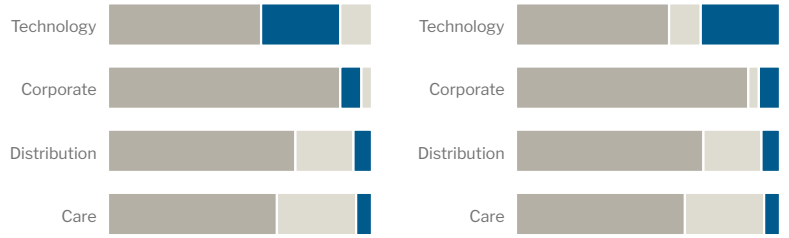
## Label in place, key in order




**Try to avoid**
A bottom legend in a different order.

**Recommended**
Values inside, key top-left in stack order.

## One order in every bar




**Try to avoid**
Segments re-sorted inside each bar.

**Recommended**
The same order everywhere.

PART-TO-WHOLE

# **Waterfall** chart

A waterfall shows how a starting total becomes an ending total, one driver at a time.

It is a stacked column chart with an invisible spacer series that floats each driver at the running total. Use it for labor cost bridges, headcount bridges and budget-to-actual walks.

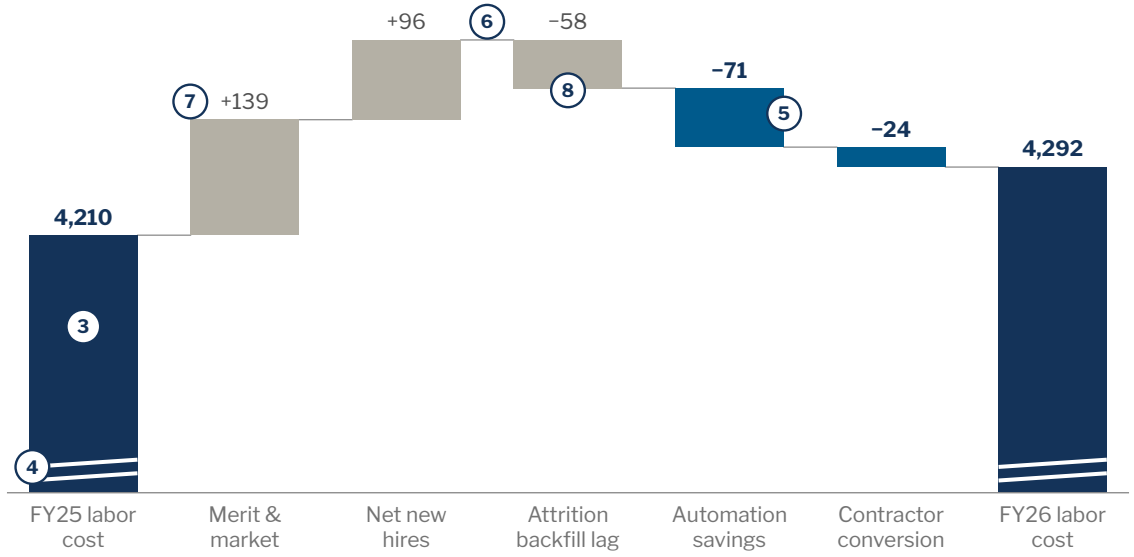
### McKesson spec

- Totals in navy; drivers grey; the drivers the headline names in blue
- Thin light connectors between bars
- Signed labels on drivers (true minus, explicit +); bold totals
- If the axis starts above zero, mark the break and say so

1. **Insight headline**
   Sizes what the named drivers did to the total.
2. **Subtitle**
   From and to, with the unit.
3. **Totals in navy**
   Start, end and any subtotal.
4. **Marked break**
   The axis starts at $3,900M; white slashes say so.
5. **The drivers that matter**
   In blue; label bold.
6. **Thin connectors**
   Light grey, so the bars stay dominant.
7. **Signed labels**
   +139, −58: direction without color.
8. **Grey context**
   Every other driver, in logical order.
9. **Footnote**
   States the axis break.

## Automation and contractor conversion offset ~40% of FY26 labor cost growth (1)

Labor cost bridge, FY25 to FY26 ($M) (2)




Value axis starts at $3,900M (break marks on the totals) so the drivers are readable.

Source: Finance labor cost model (synthetic data for illustration), FY26 (9)

PART-TO-WHOLE

# **Waterfall** variants

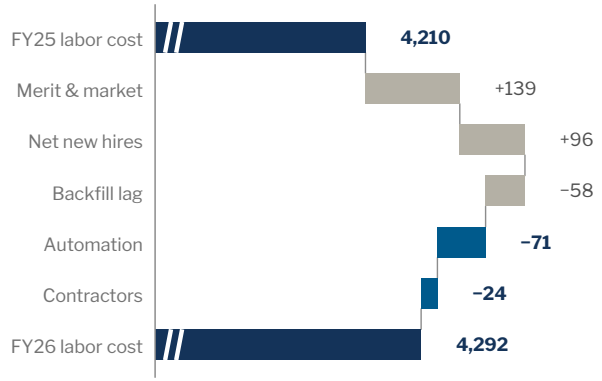
The same build, turned or recolored for the job.

**Color by direction only when direction is the point.** In a headcount bridge, additions in blue and losses in orange read instantly, because orange means "worse". In a cost bridge the default stays grey with a blue highlight.

Excel's built-in Waterfall chart is vertical only. For horizontal waterfalls or per-bar colors, build a stacked bar with a transparent spacer series (see the cookbook).

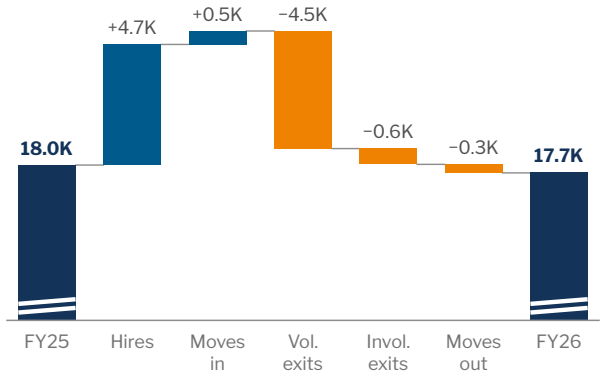
### Horizontal waterfall

Long driver names, narrow layouts and phones.



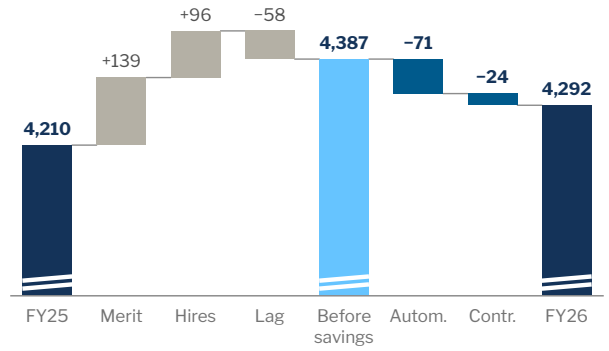

### Colored by direction

Frontline headcount bridge: voluntary exits equal 97% of external hires.



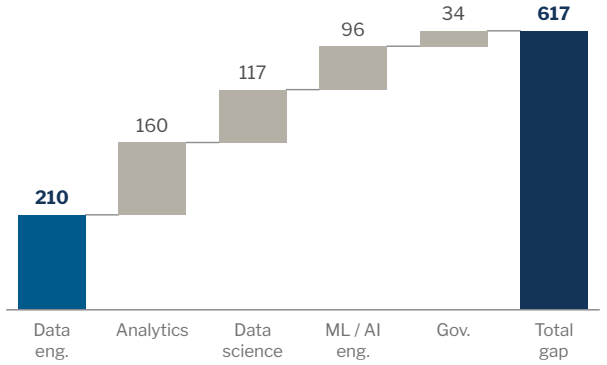

### With a subtotal

A light-blue subtotal separates cost growth from the savings that offset it.




### Distribution waterfall

Contributions to one total instead of steps in time: data engineering is 34% of the FY30 gap.



PART-TO-WHOLE

# **Waterfall** best practice

The bars carry the story; everything else stays quiet.

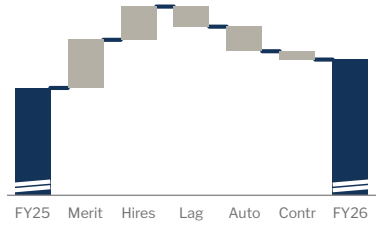
**Thin connectors.** Heavy dark connectors compete with the bars. Use hairlines in light grey.

**No red and green.** They fail for color-blind readers and aren't McKesson colors. Grey drivers with a blue highlight, or blue and orange when direction is the story.

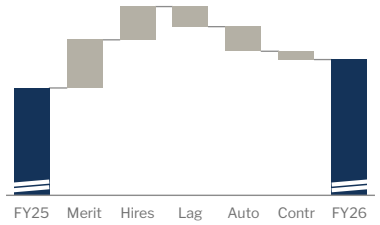
**Mark a broken axis.** Totals often dwarf the drivers. Starting the axis above zero is allowed here, and only here, if the break is drawn and footnoted.

**Group the small steps.** Keep the drivers the story needs; fold the rest together.

## Connector lines

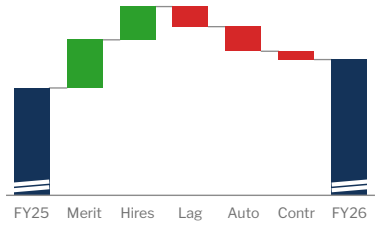



**Try to avoid**
Heavy, dark connectors.

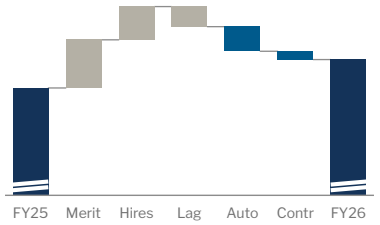



**Recommended**
Thin, light grey.

## Coloring the bars

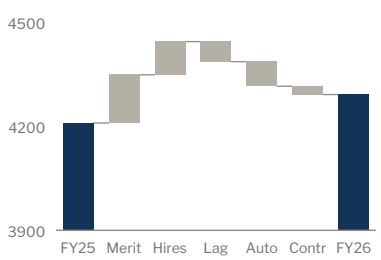



**Try to avoid**
Green up, red down.

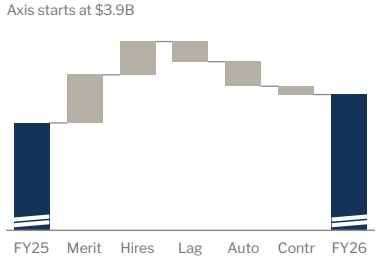



**Recommended**
Grey drivers; the named ones blue.

## A broken axis

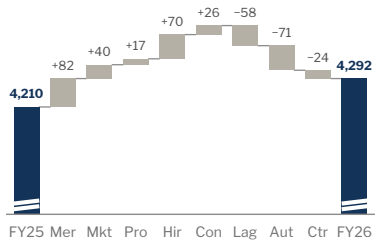



**Try to avoid**
Cut totals with no warning.

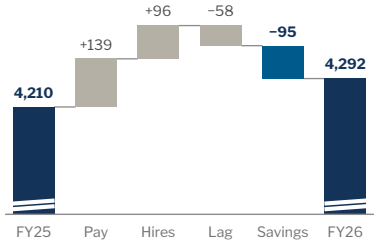



**Recommended**
Break marks and a note.

## Group the small steps




**Try to avoid**
Ten narrow drivers to decode.




**Recommended**
Five steps, savings combined.

PART-TO-WHOLE

# Treemap

Nested rectangles whose area is proportional to value. A treemap shows many parts, or a hierarchy, in one compact block.

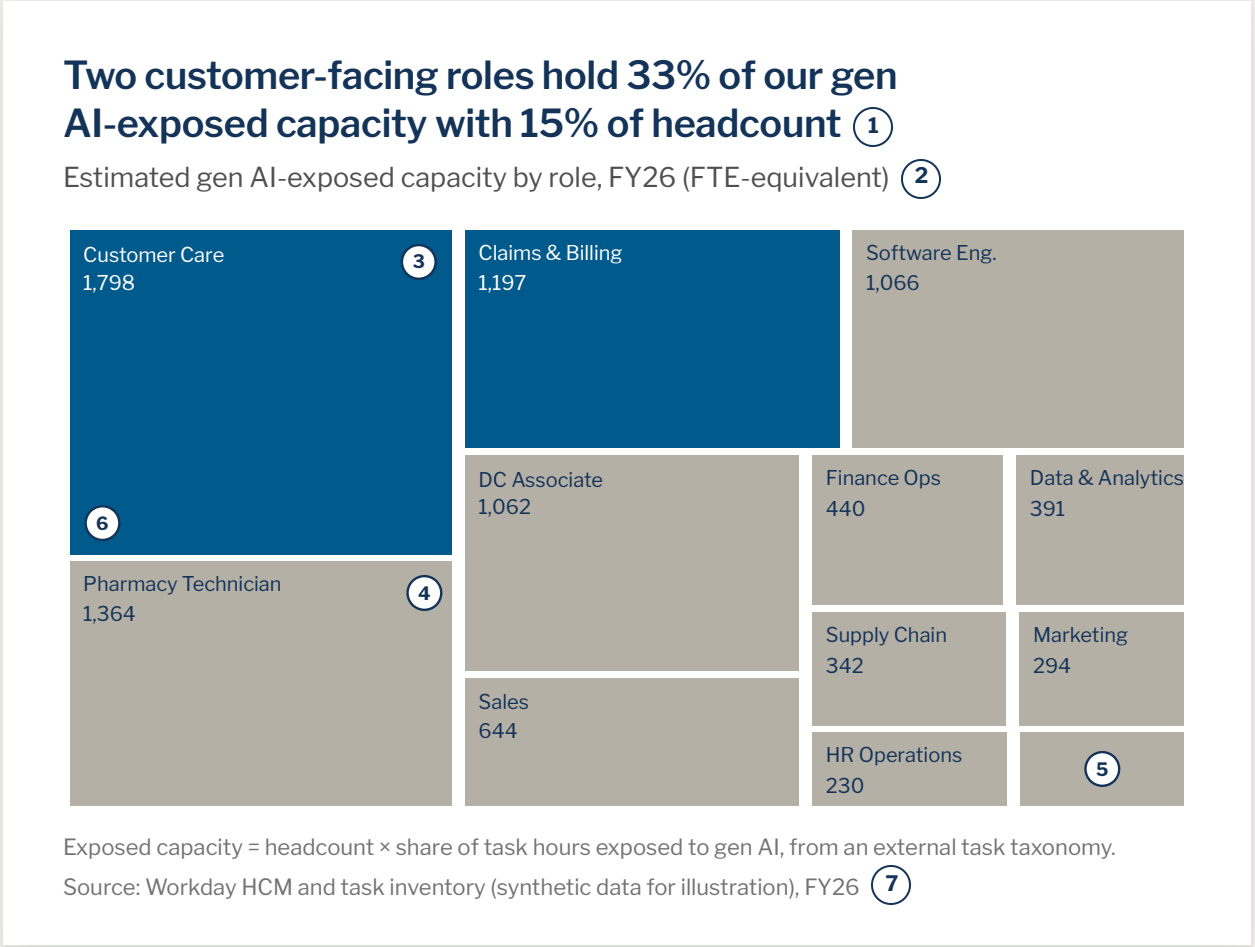
Use it when there are too many parts for a donut or a stack (ten to thirty roles, sites or skills) and the point is which few dominate.

Don't use it when readers need to compare similar sizes precisely, or there are only a handful of parts: a sorted bar is clearer.

### McKesson spec

- Squarified layout, largest top-left
- Thin, even white gutters
- Grey tiles; the ones the headline names in blue
- Name and value inside each tile that fits

1. **Insight headline**
   Share of the whole held by the named few.
2. **Subtitle**
   What area measures, with the unit.
3. **Focus in blue**
   The two roles the headline names.
4. **Grey context**
   Every other role, the same grey.
5. **Smallest last**
   Bottom-right. Tiles too small for a label get none; list them in a footnote if they matter.
6. **Largest first**
   Top-left, flowing to the smallest; labels top-left inside each tile.
7. **Footnote and source**
   Define the measure.



PART-TO-WHOLE

# **Treemap** best practice

A treemap works when it reads as one shape that flows from large to small.

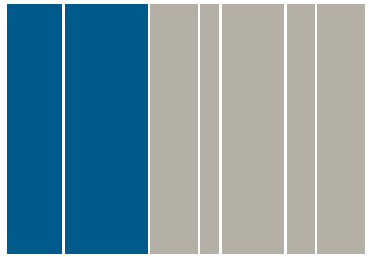
**Largest to smallest.** Don't arrange tiles alphabetically or by an org grouping that breaks the flow.

**Labels inside.** No label bands or boxes over the tiles; they distort the areas. If a tile can't hold its label, leave it unlabelled or label it outside.

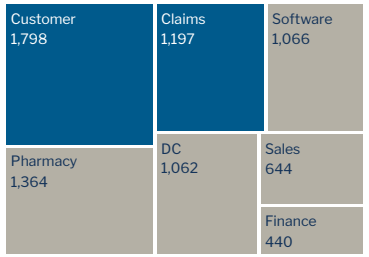
**Even padding.** Uneven gaps make similar tiles look different.

**Few parts? Use a bar.** Areas are harder to compare than lengths.

## Arrange by size

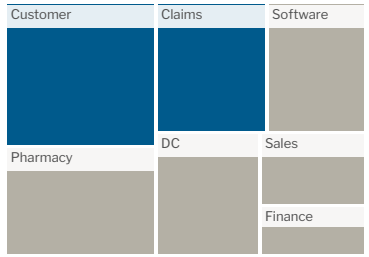


**Try to avoid**
Alphabetical strips.

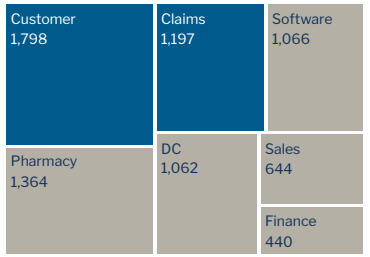



**Recommended**
Largest top-left.

## Labelling




**Try to avoid**
Label bands over the tiles.




**Recommended**
Inside, top-left.

## Padding

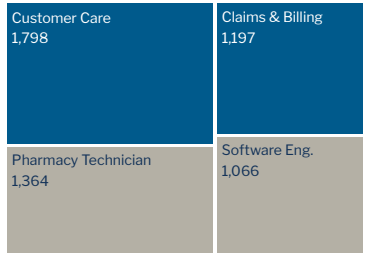


**Try to avoid**
Uneven gaps.

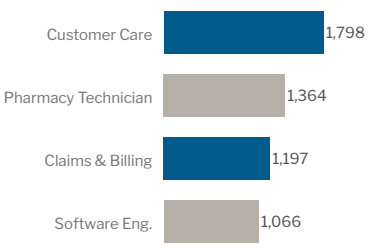


**Recommended**
Thin, even gutters.

## Few parts




**Try to avoid**
Four similar tiles.




**Recommended**
A sorted bar ranks them.

PART-TO-WHOLE

# **Donut** and pie

A donut shows one whole split into a few parts. Prefer it to a pie: readers follow the arc length around the ring, and the center holds the total.

Use it for one whole and at most five parts. To compare several wholes, use 100% stacked bars; for more parts, a bar or treemap.

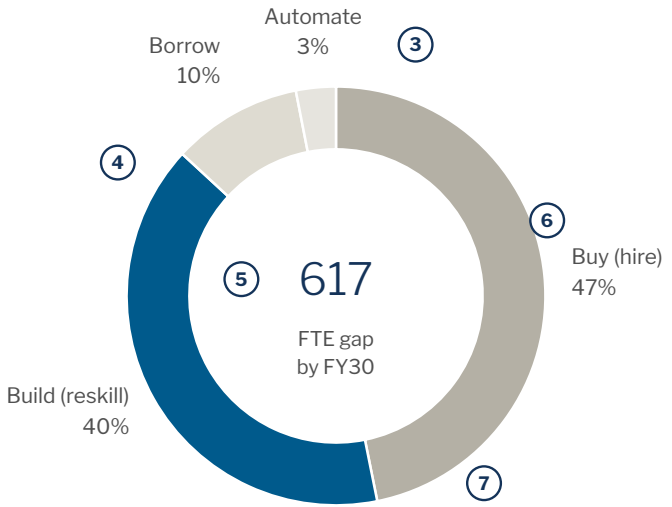
### McKesson spec

- Five segments at most; "Other" last
- Largest first, clockwise from 12 o'clock
- Hole about 60% of the diameter; white separators
- Labels and values outside; no legend

1. **Insight headline**
   The share, and what it means for the plan.
2. **Subtitle**
   The whole, the unit, the year.
3. **Start at 12**
   Largest first, then clockwise.
4. **The part that matters**
   In blue; the rest grey.
5. **Total in the center**
   Light weight, large.
6. **Labels outside**
   Name and share, beside each segment.
7. **White separators**
   Thin gaps keep segments distinct.
8. **Footnote and source**
   Define the parts.

## Reskilling can close 40% of the FY30 Data & AI gap; hiring still has to cover nearly half (1)

Planned sources to close the Data & AI capability gap, FY30 (% of 617 FTE) (2)




Base scenario. Build = reskilling current employees; buy = external hires; borrow = contractors and partners; automate = roles removed by automation.

Source: Strategic workforce planning model (synthetic data for illustration), Mar 2026 (8)

PART-TO-WHOLE

# **Donut** best practice

People judge length faster than angle or area, which is why the donut beats the pie.

**Donut over pie.** The ring turns angles into arc lengths, and the hole holds the total.

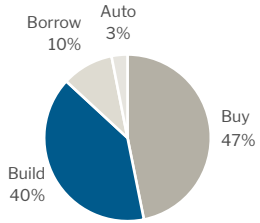
**Labels outside.** Squeezed labels on the ring are hard to read; put names and values outside.

**Few segments.** Group the smallest into "Other".

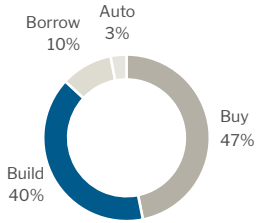
**Largest first, clockwise from 12.** Don't mix small and large at random.

Progress rings and dials for a single percentage are in Single number.

## Donut over pie

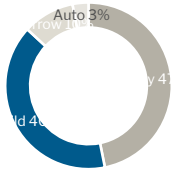



**Try to avoid**
Angles and areas.

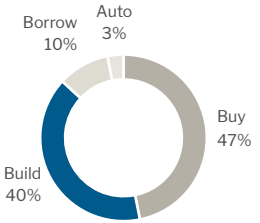



**Recommended**
Arc lengths on a ring.

## Labelling




**Try to avoid**
Labels squeezed onto the ring.

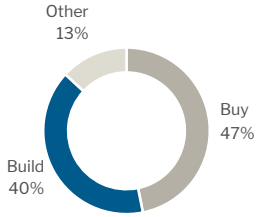



**Recommended**
Outside, with the share.

## Many small segments

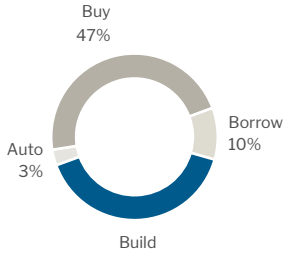


**Try to avoid**
Nine slices, nine colors.

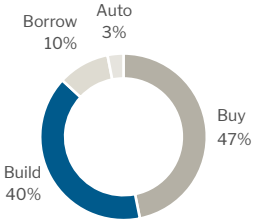



**Recommended**
The story, plus "Other".

## Order of segments




**Try to avoid**
No order, odd start.




**Recommended**
Largest first from 12.

PART-TO-WHOLE

# **Build** part-to-whole charts

Every tool can make these natively. The settings below are the ones that matter; tested code and click-by-click steps are on the cookbook pages.

stacked-bar.md · waterfall.md · treemap.md · donut-pie.md

### Before you send it

- [ ] Parts sum to the whole (100%, or the end total)
- [ ] Five parts at most; "Other" last
- [ ] One part in color, matching the headline
- [ ] Waterfall axis break marked and footnoted
- [ ] No group under 10 people, in any part

| | |
|---|---|
| **PowerPoint and Excel** | **Stacked:** Stacked or 100% Stacked Bar, Gap Width 50%, Overlap 100%, white 1.5 pt borders; move the story series first or last in Select Data. **Waterfall:** Insert › Waterfall, then right-click each total › *Set as Total*; drivers grey, totals navy. **Treemap:** Insert › Hierarchy › Treemap; labels *Overlapping*, not Banner. **Donut:** Doughnut, angle of first slice 0°, hole 62%, data sorted descending. |
| **Power BI** | Native 100% stacked bar, Waterfall, Treemap and Donut visuals all take the theme. Set the story color per series or with a field-value color measure; legend top-left or off; data labels on. The native waterfall can't color single bars or break the axis: use a stacked column with a transparent spacer measure when you need that. |
| **Python** | `ax.barh(..., left=running_total, edgecolor="white")` for stacks; the waterfall, squarified treemap and donut recipes in the cookbook use only matplotlib and `mckviz`. Plotly has native `go.Waterfall`, `go.Treemap` and `go.Pie(hole=0.6)`. |
| **HTML** | Plotly with `MCK_PLOTLY_TEMPLATE`: `barmode: "stack"`, `waterfall` and `treemap` traces. Read colors from `var(--mck-chart-focus)` and `var(--mck-chart-context)`. |
| **R** | `geom_col(position = "fill")` for 100% stacks; `treemapify::geom_treemap()` for treemaps; waterfalls with `geom_rect()`. Colors from `mck_highlight()`. |

```
colors = mv.highlight(parts, focus="Contingent")          # one part blue, rest grey
mv.lint(fig)    # flags > 6 colors, legends, and bars that don't start at zero
```

CHANGE OVER TIME

# **Line** chart

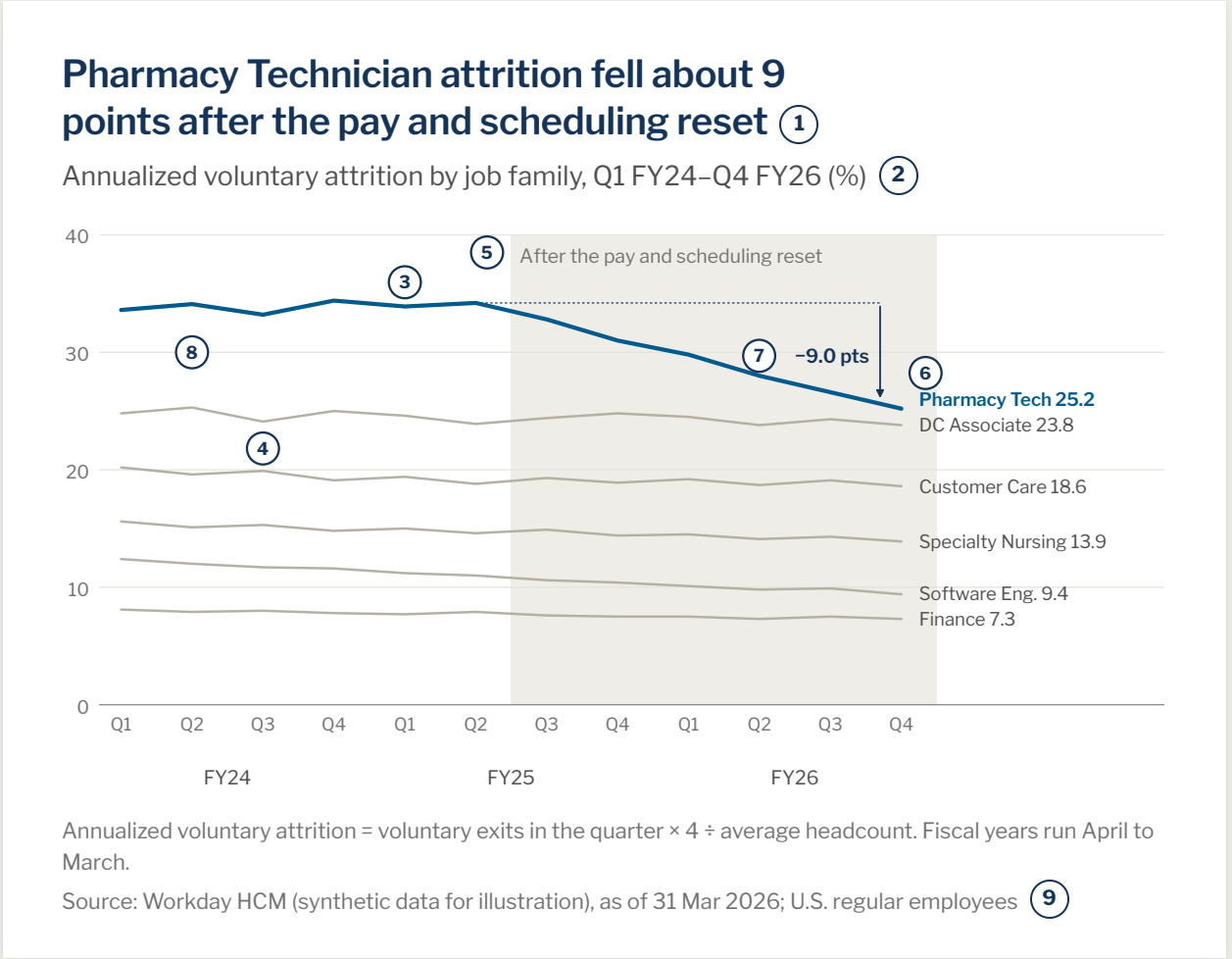
Change-over-time charts show how something moves, and when it changed, at any time scale from weeks to decades.

A line connects values at equal time intervals, so the eye reads the trend rather than the individual amounts. Use it for monthly or quarterly attrition, hiring velocity, or a forecast.

### McKesson spec

- The series the headline names in blue; others grey
- Data lines about twice the gridline weight
- Names at the line ends, not in a legend
- Shade the period that explains the change

1. **Insight headline**
   What changed, by how much, and why.
2. **Subtitle**
   Metric, unit, scope and period.
3. **Focus line**
   Blue and heavier than the rest.
4. **Context lines**
   Grey and thinner: there to compare against.
5. **The period that matters**
   Lightly shaded and named.
6. **Direct labels**
   Names and latest values at the line ends, nudged apart.
7. **The change, sized**
   A short callout carries the headline number.
8. **Light scaffolding**
   A few thin gridlines; the unit lives in the subtitle.
9. **Footnote and source**
   Define the metric.



CHANGE OVER TIME

# Line chart variants

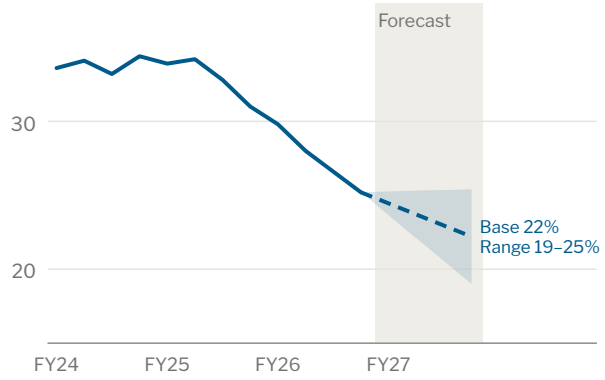
The same rules hold: one line in blue, names at the ends, light scaffolding.

### When a line is the wrong chart

- Only two to five periods, and the amounts matter: **columns**
- How a total splits over time: **stacked area**
- Discrete events of different sizes: a **bubble timeline**
- Many series that tangle: **small multiples**
- Two measures in different units: two aligned charts, never a **dual axis**

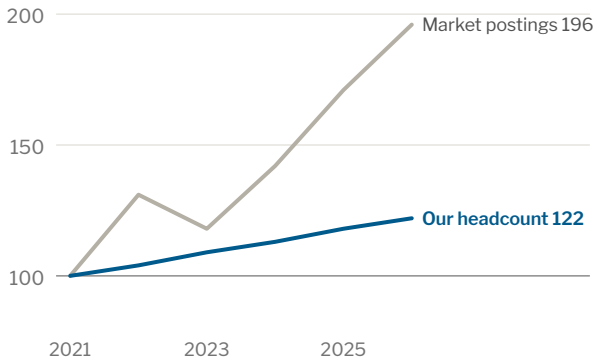
### Forecast with a range

Dashed beyond the last actual, inside a band that widens with time. Label it as an estimate.



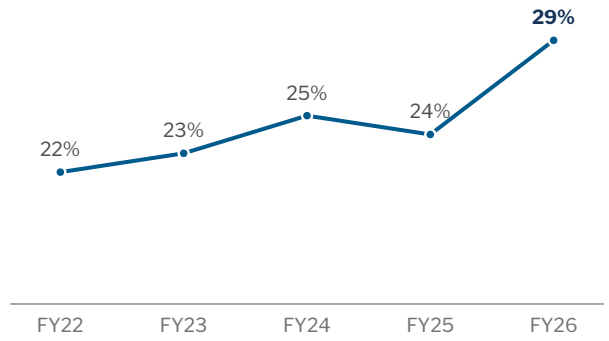

### Indexed to 100

Compares growth when the units or sizes differ: market Data & AI postings reached 196, our headcount 122.



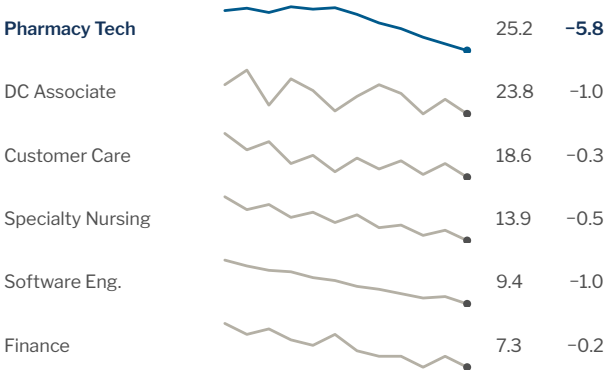

### Few points, every value labelled

With five annual values, label each one and drop the axis.




### Sparklines

One small line per row in a table or dashboard: shape, latest value, change. Worse in orange-brown.



CHANGE OVER TIME

# **Line chart** best practice

Four rules that make trends readable.

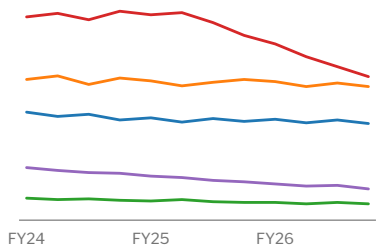
**Highlight one line.** A color per series doesn't show the trend; it makes the reader decode. Grey the background, color the message.

**Lines for trends.** Stacked columns hide how each part changes. When the trend is the point, draw lines.

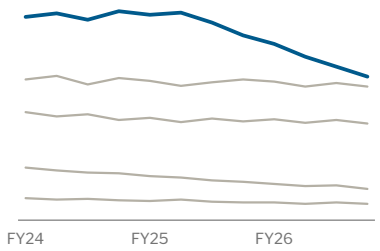
**No fill without a zero.** Shading under a line turns it into an area chart, which implies a zero baseline.

**Names at the line ends.** A legend above the chart breaks the reading order.

## Highlights

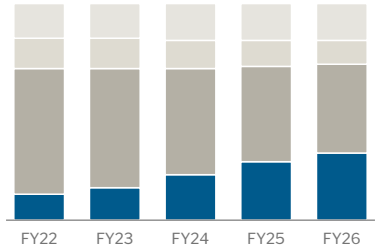



**Try to avoid**
Every line its own color.

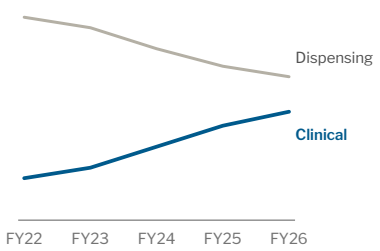



**Recommended**
Grey context, one blue line.

## Showing trends

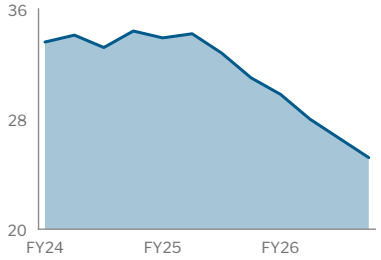



**Try to avoid**
Stacks hide each part's change.

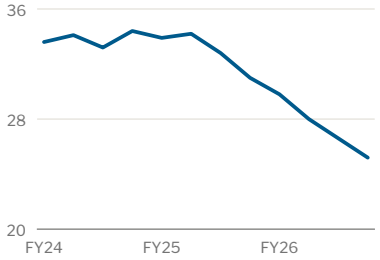



**Recommended**
Lines show the crossover.

## Shading lines

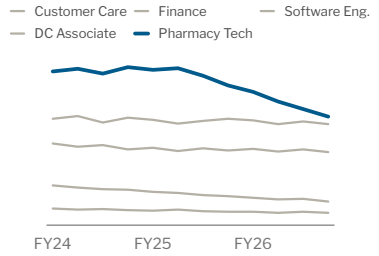



**Try to avoid**
A filled area on a 20–36 axis.

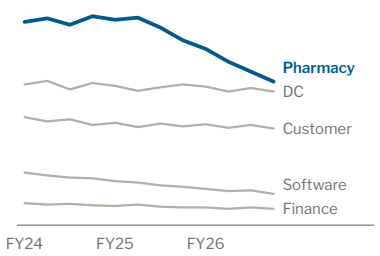



**Recommended**
A plain line.

## Labelling lines




**Try to avoid**
A legend to decode.




**Recommended**
Names where the lines end.

CHANGE OVER TIME

# **Line chart** best practice, continued

Four more, from HR and workforce data in particular.

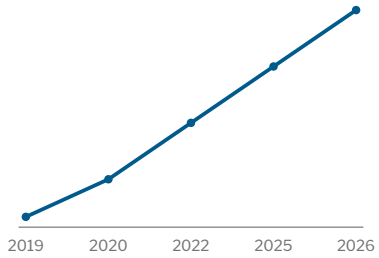
**Honest time spacing.** Missing years still take up space. Use a real date axis, not evenly spaced labels.

**Show gaps as gaps.** A system change or survey break is not a straight line. Leave the gap and say why.

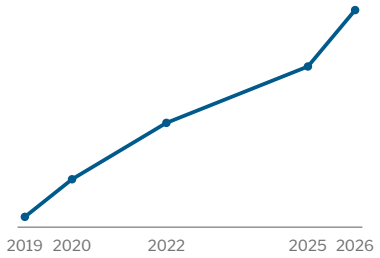
**Weight with intent.** Data lines about twice the gridlines; gridlines light, or gone.

**Too many lines? Split them.** More than four or five crossing lines become spaghetti. Use small multiples.

## Even time intervals

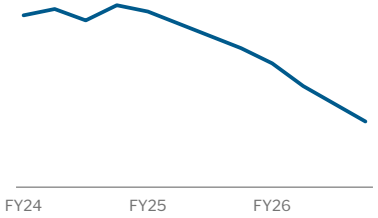



**Try to avoid**
Missing years squeezed out.

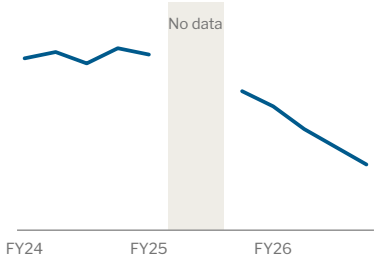



**Recommended**
True spacing; the slope is honest.

## Missing data

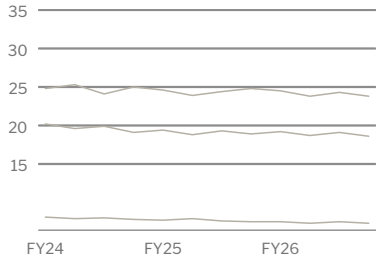



**Try to avoid**
Drawing across the gap.

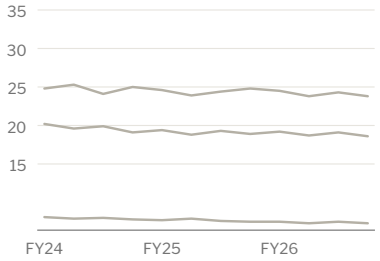



**Recommended**
A visible, labelled gap.

## Line weight

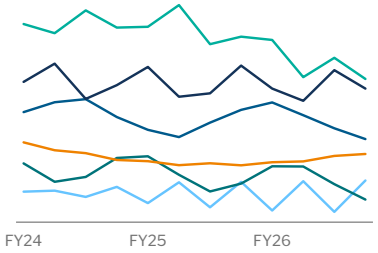



**Try to avoid**
Gridlines outweigh the data.

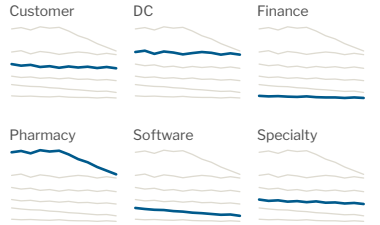



**Recommended**
Data lines twice the grid.

## Too many lines




**Try to avoid**
Six crossing colors.




**Recommended**
Small multiples.

CHANGE OVER TIME

# **Stacked** area chart

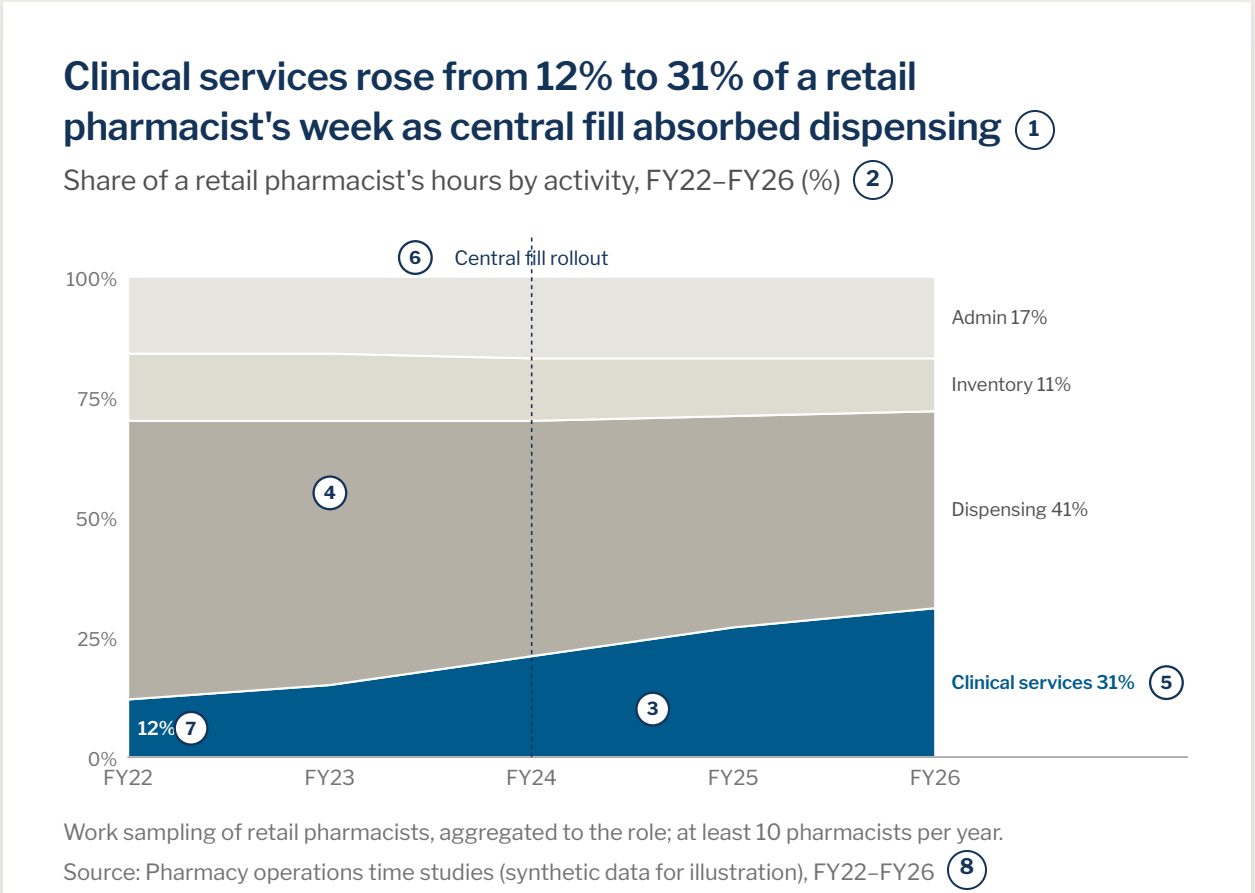
Layers stacked on each other show how a total splits over time, and how each part's share shifts.

Use it for a few parts that change gradually: how a role's hours divide between activities, or how the workforce mix moves. The layer on the baseline is the only one that reads precisely, so put the story there.

### McKesson spec

- Four or five layers at most
- The story layer on the baseline, in blue; the rest ordered greys
- Labels inside or at the right edge, not in a legend
- Mark the event that explains the shift

1. **Insight headline**
   Start, end and the reason.
2. **Subtitle**
   Whose time, what unit, which years.
3. **Story on the baseline**
   Blue, so its growth reads directly.
4. **Grey context**
   Ordered from dark to light.
5. **Labels at the end**
   Name and latest share beside each layer.
6. **The event, marked**
   A thin dashed line and a short name.
7. **Where it started**
   The first value, so the change is explicit.
8. **Footnote and source**
   How time was measured.



CHANGE OVER TIME

# **Stacked area** best practice

Areas are heavy marks. Keep them few, ordered and calm.

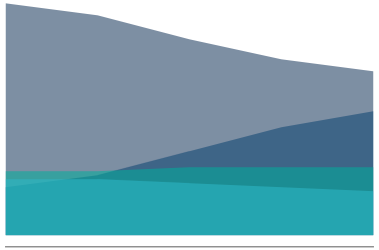
**Stack, don't overlap.** Areas drawn over each other hide what's behind. Stack them, or use lines.

**Order the colors.** Random colors make neighbors clash. Story layer blue on the baseline; the rest in ordered greys.

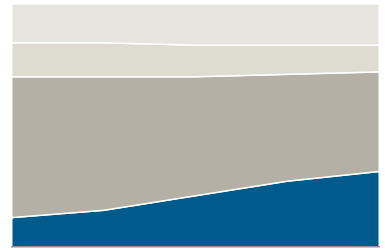
**Volatile on top.** A jumpy layer at the bottom makes every layer above it jump. Keep steady layers low.

**Few layers.** Four or five at most; fold the rest into "Other".

### Overlapping areas

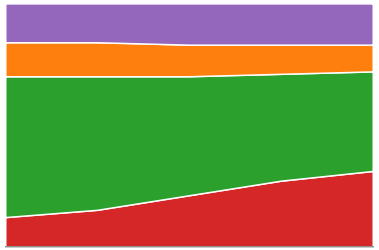


**Try to avoid**
Translucent layers over each other.

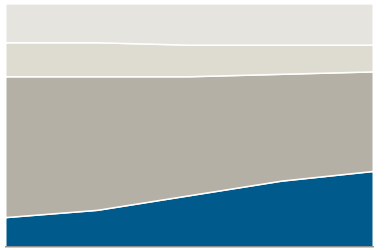


**Recommended**
Stacked layers.

### Color order



**Try to avoid**
Unrelated bright colors.



**Recommended**
Blue story, ordered greys.

### High variability

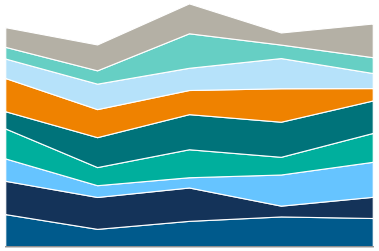


**Try to avoid**
The volatile layer at the bottom.



**Recommended**
Steady layers low, volatile on top.

### Number of layers



**Try to avoid**
Nine thin layers.



**Recommended**
Three: story, comparison, rest.

CHANGE OVER TIME

# **Bubble** timeline

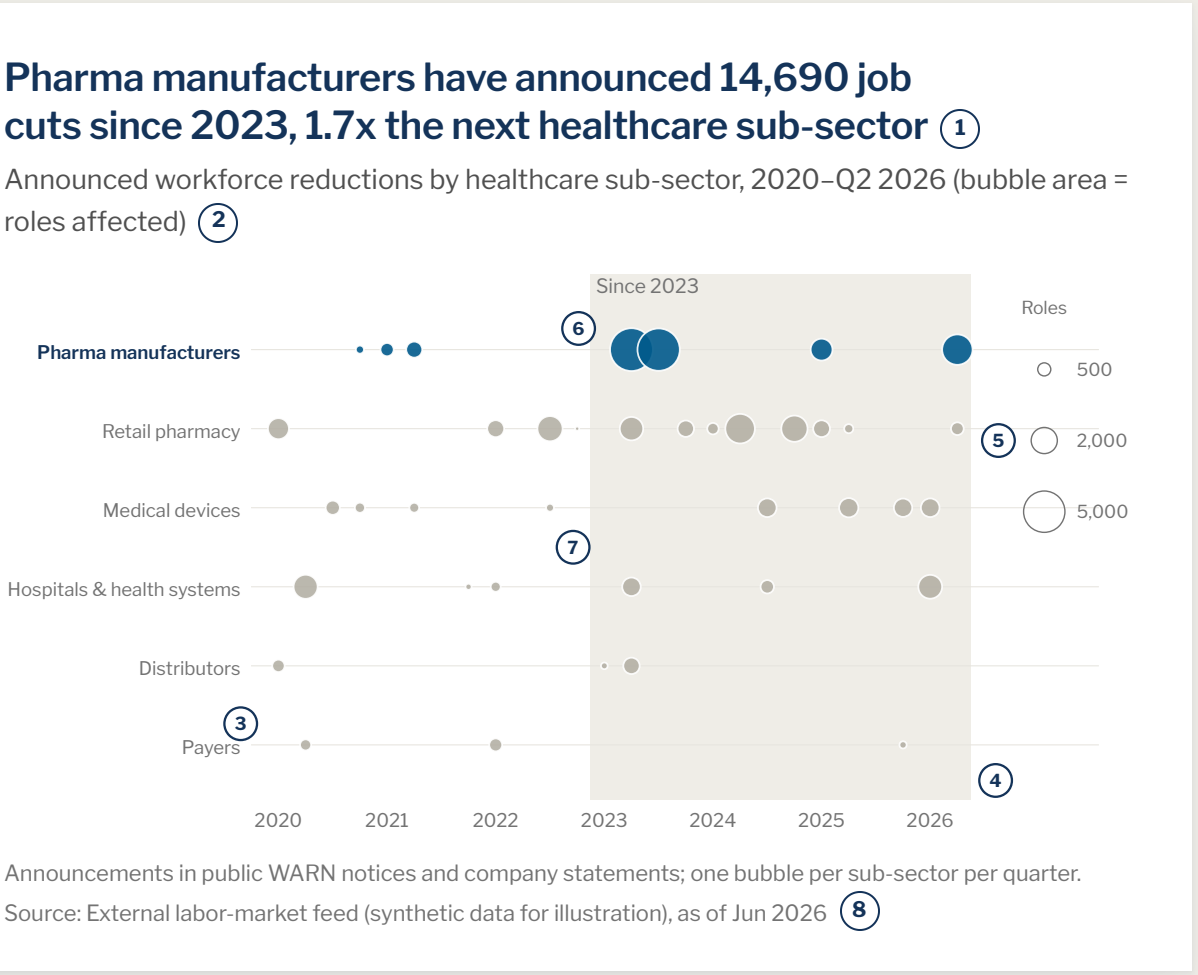
Events placed on a time axis, with bubble area showing size. It combines a timeline with a proportional-area chart.

Use it for discrete events of different sizes: announced layoffs in the talent market, reorganizations, hiring waves. Each category gets its own row.

### McKesson spec

- Bubble area, not diameter, proportional to value
- One grey for all rows; the headline's row in blue
- Largest bubbles drawn first, with white outlines
- A size key with two or three reference circles

1. **Insight headline**
   Who stands out, by how much.
2. **Subtitle**
   Says what bubble area means.
3. **One row per category**
   Sorted by total, largest on top.
4. **Time along the bottom**
   Even intervals, years labelled.
5. **Size key**
   Reference circles; area ∝ value.
6. **Focus row**
   Blue; white outlines keep overlaps readable.
7. **The period that matters**
   Shaded and named.
8. **Footnote and source**
   What counts as an event.



CHANGE OVER TIME

# **Bubble timeline** best practice

The rows already separate the categories; let size do the rest.

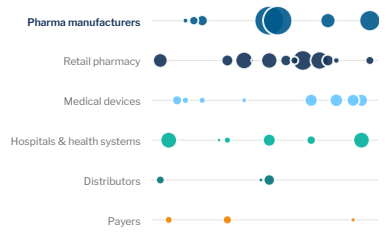
**One color.** Coloring each row repeats what the rows already say. Use grey, and blue for the headline's row.

**Don't hide small bubbles.** Draw the largest first, with white outlines.

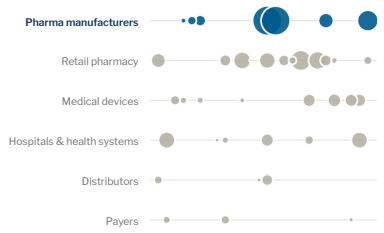
**Sort the rows.** By total, or by a meaningful order, not alphabetically.

**Show the scale.** Without a size key, readers can only say "bigger".

## Coloring bubbles

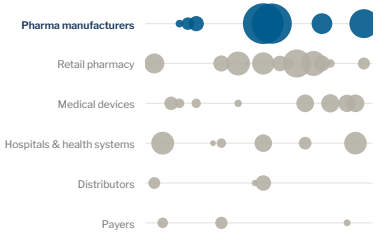



**Try to avoid**
A color per row.

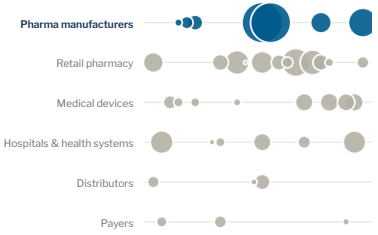



**Recommended**
Grey, one row blue.

## Overlapping bubbles

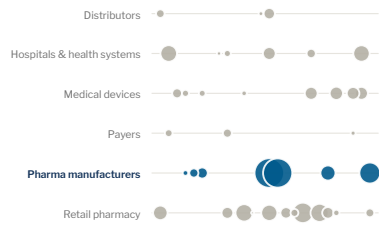



**Try to avoid**
Big bubbles on top.

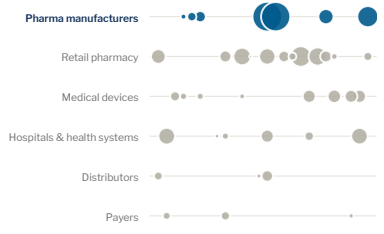



**Recommended**
Big behind, white outlines.

## Row order

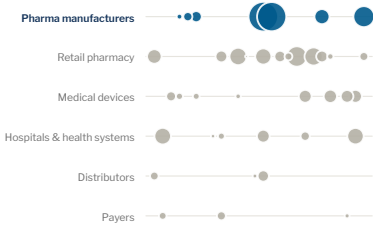



**Try to avoid**
Alphabetical rows.

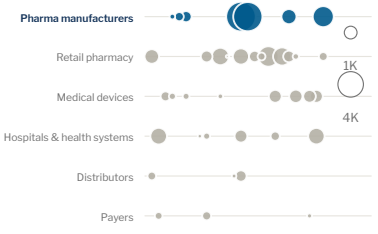



**Recommended**
Sorted by total.

## Size key




**Try to avoid**
No way to read size.




**Recommended**
Two reference circles.

CHANGE OVER TIME

# **Small** multiples

The same small chart repeated for each group, on one shared scale. The eye compares shapes across panels instead of untangling crossing lines.

Use it when six or more groups each need their own trend: business units, regions, job families. In each panel, the panel's own line is blue and the others sit faintly behind it.

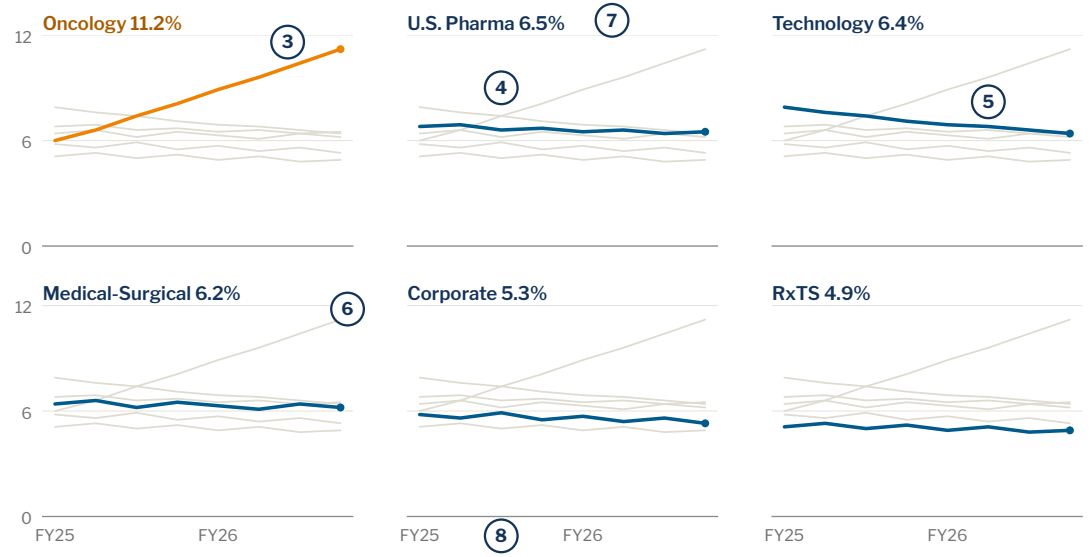
### McKesson spec

- One shared scale, stated once
- Panels ordered by value or a natural order
- Panel title = name + latest value
- Axis labels only on the outer edge

1. **Insight headline**
   The one panel that differs.
2. **Subtitle**
   Metric, unit, period, once for all panels.
3. **The exception**
   In orange, because it is getting worse.
4. **Its own line in blue**
   The other units faint behind it, for context.
5. **Ordered by value**
   Highest latest value first.
6. **Shared scale**
   0–12% in every panel.
7. **Title carries the number**
   Name and latest value.
8. **Outer axes only**
   Labels on the left column and bottom row.
9. **Footnote and source**
   Define the metric.

## Oncology regretted attrition nearly doubled in two years while every other business unit stayed within 2 points (1)

Regretted attrition by business unit, Q1 FY25–Q4 FY26 (% of high performers, annualized) (2)




Regretted attrition = voluntary exits of employees rated "exceeds" in the last review, annualized.

Source: Workday HCM (synthetic data for illustration), as of 31 Mar 2026 (9)

CHANGE OVER TIME

# **Small multiples** best practice

Small multiples only work if every panel is read the same way.

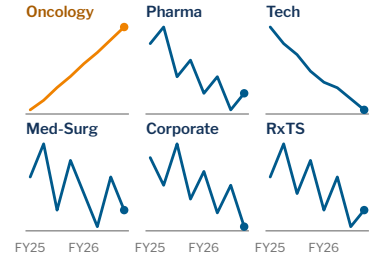
**One scale.** Independent scales make a 0.5-point wobble look like the 5-point rise.

**Context in every panel.** The other units, faint behind each line, show whether a panel is unusual.

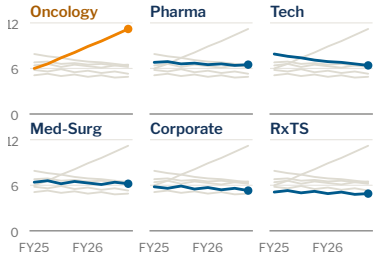
**Order with meaning.** By value, size or geography, not the alphabet.

**Label the edges only.** Repeated axes are clutter the eye reads six times.

## Shared scale

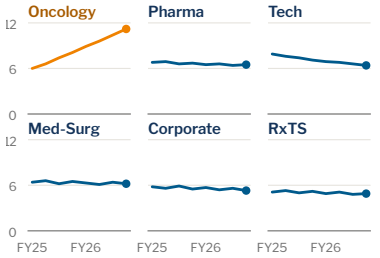



**Try to avoid**
Every panel zooms itself.

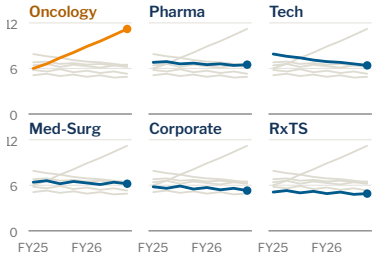



**Recommended**
One scale; the rise stands out.

## Context

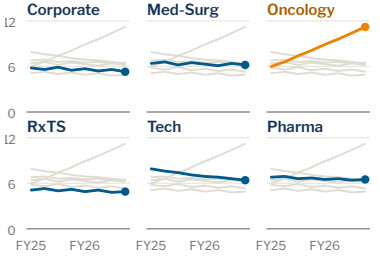



**Try to avoid**
Lines alone.

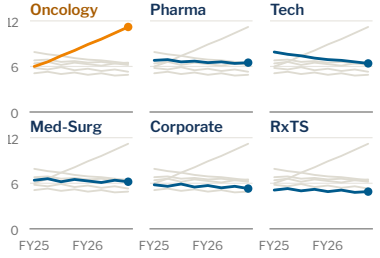



**Recommended**
The others faint behind.

## Panel order

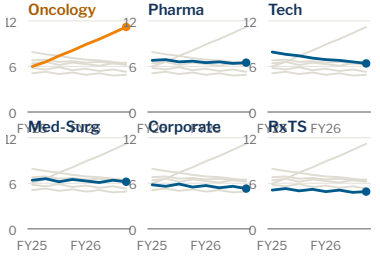



**Try to avoid**
Alphabetical.

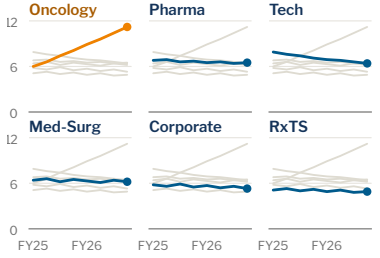



**Recommended**
By latest value.

## Axes




**Try to avoid**
Axes on every panel.




**Recommended**
Outer edge only.

CHANGE OVER TIME

# **Build** change-over-time charts

Native, editable charts in every tool. Tested code and click-by-click steps are on the cookbook pages.

line.md · stacked-area.md · bubble-timeline.md · small-multiples.md

### Before you send it

- [ ] Time left to right, at true intervals
- [ ] One line or layer in blue, matching the headline
- [ ] Names at line ends; no legend
- [ ] Forecasts dashed, labelled and with a range
- [ ] Small multiples share one scale

| | |
|---|---|
| **PowerPoint and Excel** | **Line:** Insert › Line; format the focus series 2.25 pt Focus blue, the rest 1.25 pt grey; delete the legend and add data labels to the last point only (Series Name + Value). Use a *date* axis so missing periods keep their space. **Stacked area:** Insert › 100% Stacked Area; story series first in Select Data; white 1 pt borders. **Small multiples:** build one chart, fix the axis bounds (Minimum/Maximum, not Auto), then copy it per group. |
| **Power BI** | Line chart with the focus series colored by measure; turn off the legend and use *Series labels*. Forecast bands as low/high measures shown with *Error bars* › *Shaded area*. Small multiples are built in: add the group to *Small multiples* and keep *Y-axis* › *Shared*. |
| **Python** | matplotlib with `mckviz`; `mv.label_line_ends(ax)` places and nudges end labels; `mv.shade_period(ax, start, end, label)` marks the period. |

```
for fam, s in df.items():                # one column per job family, one row per quarter
    ax.plot(x, s, label=fam, color=C.focus if fam == focus else C.context,
            lw=2.4 if fam == focus else 1.4)
mv.shade_period(ax, 6, 11, "After the reset")
mv.label_line_ends(ax, fmt="{:.1f}")      # names at the ends, nudged apart
```

| | |
|---|---|
| **HTML** | Plotly `scatter` traces with `mode: "lines"`, an annotation per line end, and a `shape` for the shaded period; `facet_col` (Plotly Express) for small multiples with shared axes. |
| **R** | `geom_line()` with `mck_highlight()` colors and `geom_text()` end labels; `geom_area()` for stacks; `facet_wrap(~ unit)` keeps a shared scale by default. |

RANKING

# **Slope** chart

Ranking charts show order: who comes first, who moved, and how far apart the items really are.

A slope chart joins each item's rank (or value) at two points in time. The eye reads the angle: steep lines are the movers, and crossings show who overtook whom. Use it for skills in job postings, sites or job families before and after a change.

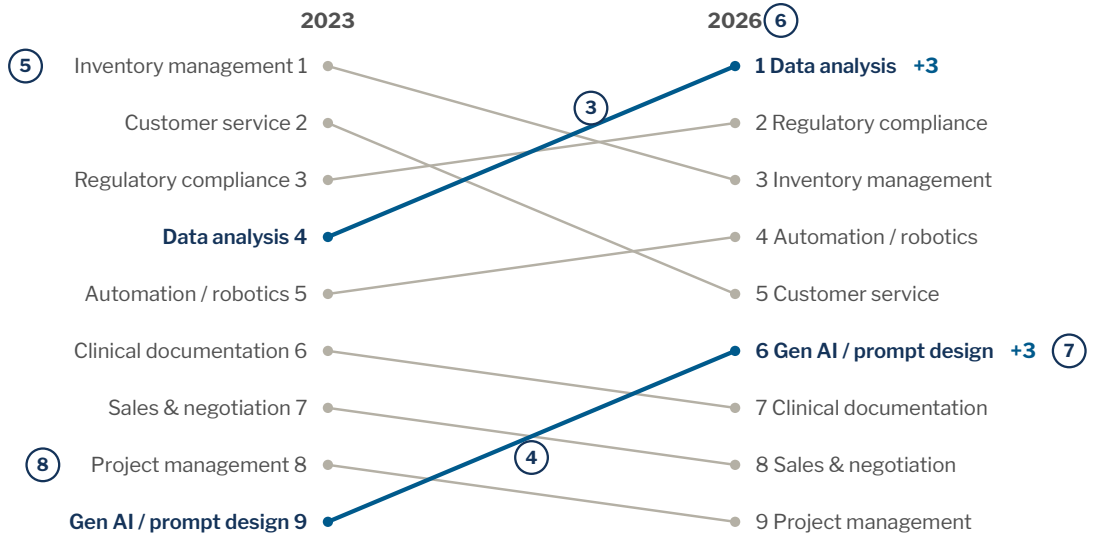
### McKesson spec

- Five to ten items; one or two movers in blue
- Names at both ends, in rank order; no legend
- Rank 1 at the top; no axis or gridlines
- Footnote what the rank is based on

1. **Insight headline**
   Names the movers and how far they moved.
2. **Subtitle**
   What is ranked, where, which two years, and what 1 means.
3. **Movers in blue**
   The two the headline names, drawn last so they sit on top.
4. **Grey context**
   Every other item, thinner.
5. **Names at both ends**
   Left in the starting order, right in the ending order.
6. **Period headers**
   Bold, above each column.
7. **The move, sized**
   A signed change beside the focus names.
8. **Rank 1 on top**
   Ranks evenly spaced; the numbers sit beside the names.
9. **Footnote and source**
   What the rank measures.

## Data analysis and gen AI are the fastest-rising skills in sector job postings, up three places each since 2023 (1)

Rank of skills in U.S. healthcare-distribution job postings, 2023 vs. 2026 (1 = most requested) (2)




Rank by share of job postings that mention the skill; the 9 most-requested skills in 2023.
Source: External job-postings feed (synthetic data for illustration), U.S. healthcare distribution, 2023 and 2026 (9)

RANKING

# **Slope and bump** variants

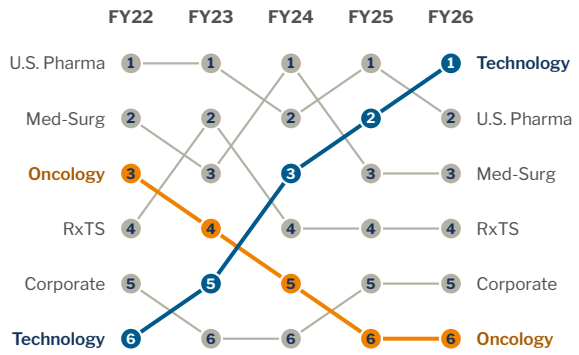
The same rules: grey context, one or two movers in color, names where the lines end.

### When a slope is the wrong chart

- The gaps matter more than the order: a **sorted bar** or a **dot plot**
- More than about six periods, or the shape between them matters: a **line chart**
- More than about ten items: show the **top N**, or use **small multiples**
- One period only: a **sorted bar**

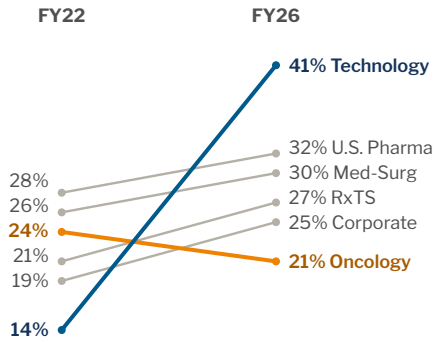
## Bump chart

Rank order over five years. Technology climbed from last to first in internal fill; Oncology slid from third to last. A fall to last is worse, so it is orange.



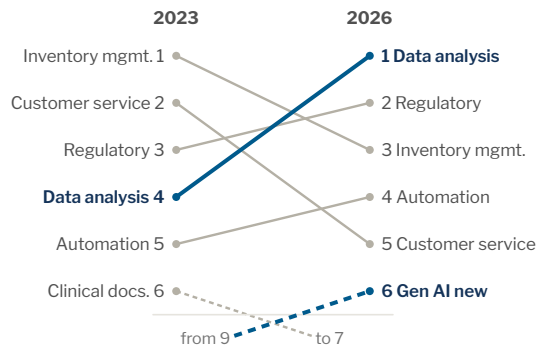

## Value slope

Values on one shared scale, when the size of the change matters: Technology went from 14% to 41%, Oncology from 24% to 21%.




## Top N, with entries and exits

“Top 6” in the subtitle, and a stub for what enters or leaves: gen AI enters from ninth; clinical documentation drops to seventh.




## Ranked list with movement

For dashboards and long lists: the current order, with the change in rank in a column of its own.

| 2026 | | | vs. 2023 |
|---|---|---|---|
| **1** | **Data analysis** | ▲ | **+3** |
| 2 | Regulatory compliance | ▲ | +1 |
| 3 | Inventory management | ▼ | −2 |
| 4 | Automation / robotics | ▲ | +1 |
| 5 | Customer service | ▼ | −3 |
| **6** | **Gen AI / prompt design** | ▲ | **+3** |
| 7 | Clinical documentation | ▼ | −1 |
| 8 | Sales & negotiation | ▼ | −1 |
| 9 | Project management | ▼ | −1 |

RANKING

# **Slope and bump** best practice

Crossing lines are the story, so make them easy to follow.

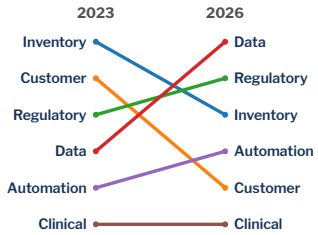
**Highlight the movers.** A color per line makes every crossing compete. Grey everything, then color the one or two the headline names.

**Names at both ends.** A legend sends the reader back and forth. Label each line where it starts and where it ends.

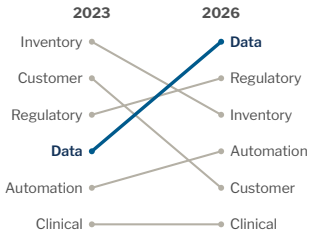
**Rank 1 on top.** A normal value axis puts first place at the bottom. Reverse it and drop the axis.

**Ranks hide gaps.** A swap of places looks like an even trade. When the size of the change matters, plot values.

## Highlights

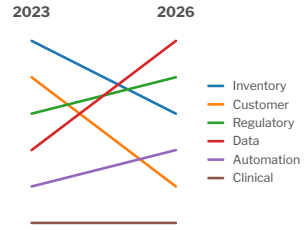



**Try to avoid**
A color for every line.

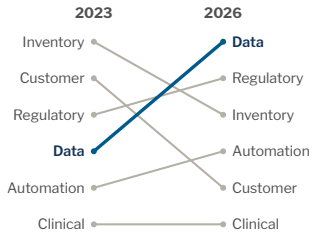



**Recommended**
Grey context, the mover in blue.

## Labelling

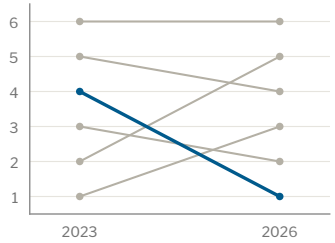



**Try to avoid**
A legend to decode.

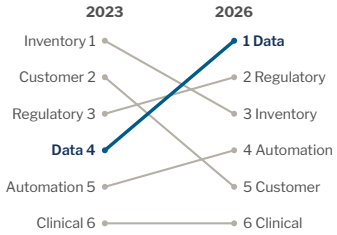



**Recommended**
Names at both ends.

## Rank order

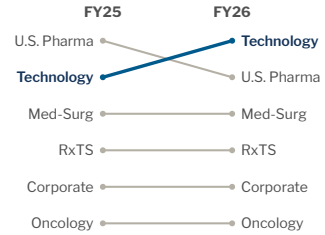



**Try to avoid**
Rank 1 at the bottom of an axis.

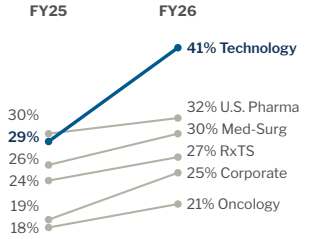



**Recommended**
Rank 1 on top, numbers by the names.

## Ranks hide gaps




**Try to avoid**
A swap that looks like an even trade.




**Recommended**
Values show who really moved.

RANKING

# **Dot plot** and dumbbell

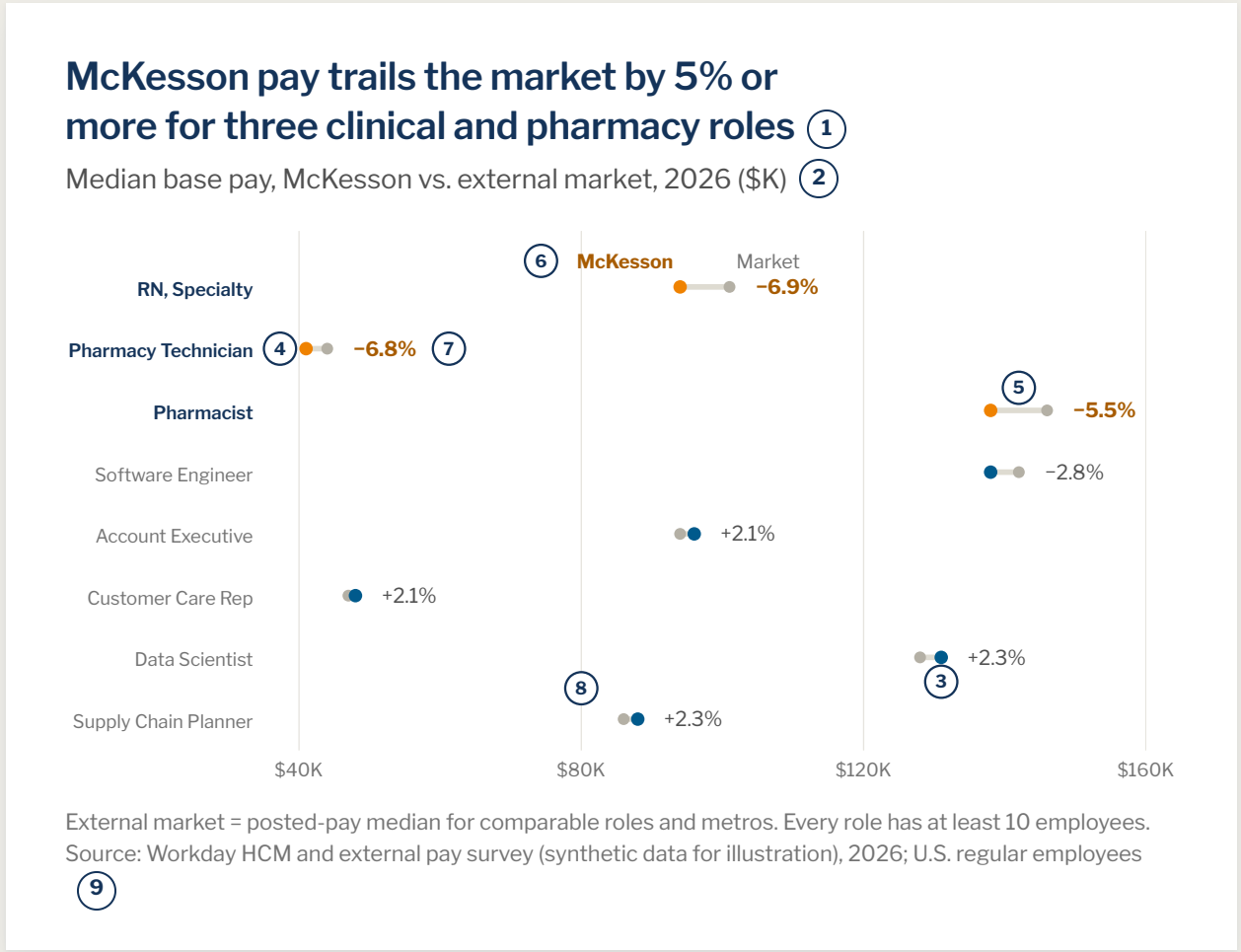
A dot encodes position, not length, so the axis can start where the data starts. A dumbbell joins two dots per row, and the eye reads the gap.

Use a dot plot for scores far from zero (engagement, compa-ratio, proficiency) and a dumbbell for us vs. market, before vs. after, or plan vs. actual.

### McKesson spec

- Comparator grey, focus blue; orange only for worse
- Light connector behind the dots
- Signed gap at the row end; sorted by the gap
- Round, non-zero range with light gridlines

1. **Insight headline**
   How many rows fall short, and by how much.
2. **Subtitle**
   Both measures, the year and the unit.
3. **Grey comparator, blue focus**
   Market grey; McKesson blue.
4. **Worse in orange**
   The roles 5% or more below market.
5. **A light connector**
   The gap is what the eye reads.
6. **Named once**
   Dot types labelled on the top row, not in a legend.
7. **Signed gap**
   At the row end, with a true minus; bold for the exceptions.
8. **Position, not length**
   A round, non-zero range and light gridlines.
9. **Footnote and source**
   Define the comparator.



RANKING

# Dot plot variants

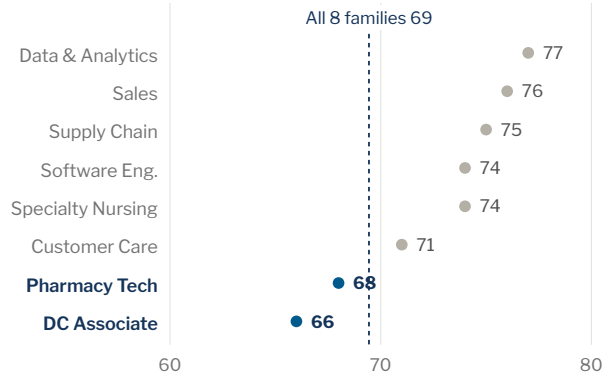
Dots carry one, two or three values per row, on any scale that suits the data.

### When a dot plot is the wrong chart

- The size from zero matters (headcount, cost): a **bar chart**
- More than two or three points in time: a **line chart**
- A change in rank: a **slope or bump chart**
- The spread across people: a **distribution**
- Gaps smaller than a dot: plot the **difference** instead

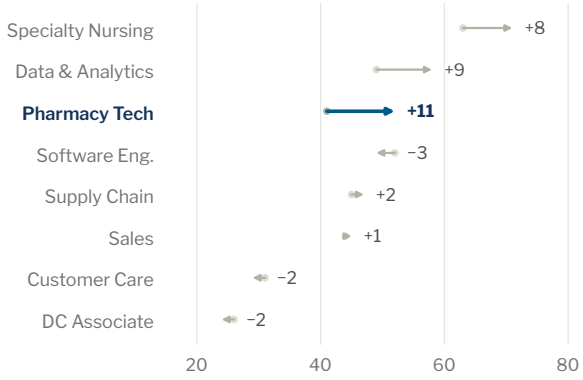
## Dot plot with a benchmark

Engagement runs 66–77, so bars would need a cut axis. The two lowest sit below the 69 average for all 8 families.



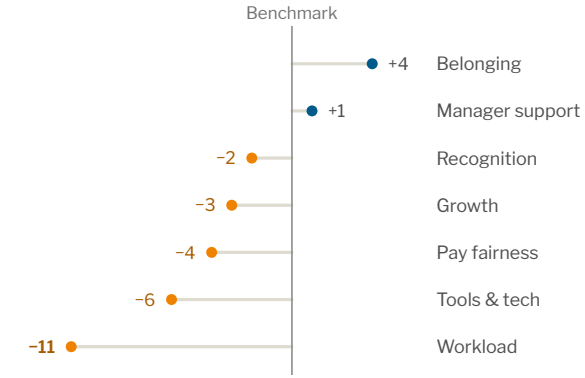

## Before and after

Arrows from FY25 to FY26 make direction explicit: pharmacy technician roles take 11 more days to fill.



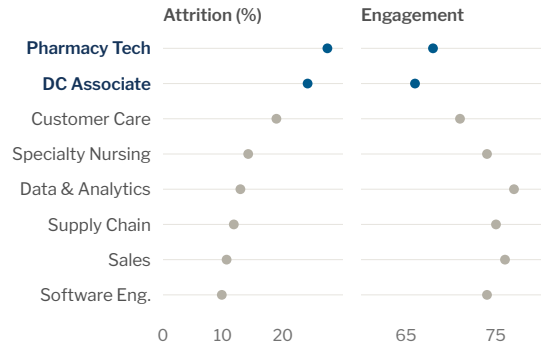

## Gaps to a benchmark

Centred on zero, below in orange: Oncology's workload score is −11 points from the survey benchmark.




## Two measures, one row order

Side-by-side panels, each on its own scale, keep the rows aligned so the pattern reads across.



RANKING

# **Dot plot** best practice

Dots are quiet marks. These four rules keep the comparison obvious.

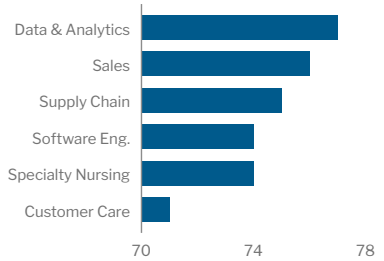
**Dots, not cut bars.** When values sit far from zero, switch to dots. Never truncate a bar.

**Connect the pair.** A light connector shows the gap; name the two dot types once, on the top row.

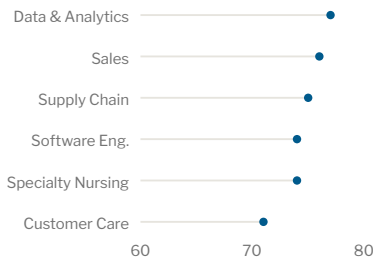
**Grey comparator.** Two strong colors leave the reader asking which one is us.

**Sort by the gap.** Alphabetical order hides the story. Put the largest shortfall first and write the gap at the row end.

## Values far from zero

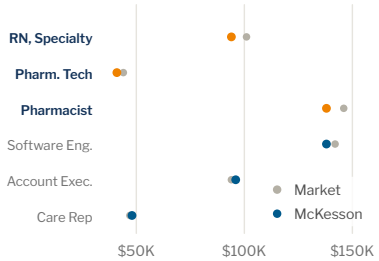



**Try to avoid**
Bars on an axis cut at 70.

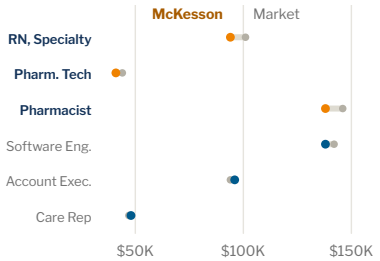



**Recommended**
Dots on an honest range.

## Connect the pair

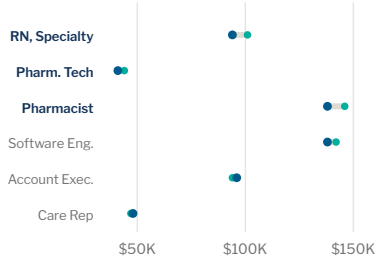



**Try to avoid**
Loose dots and a legend.

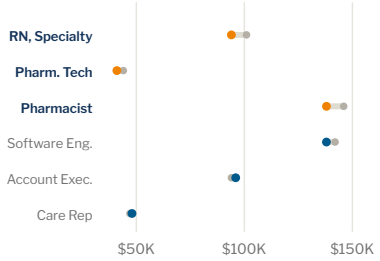



**Recommended**
A connector; types named once.

## Colors

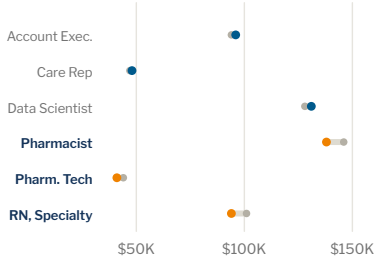



**Try to avoid**
Two equally strong colors.

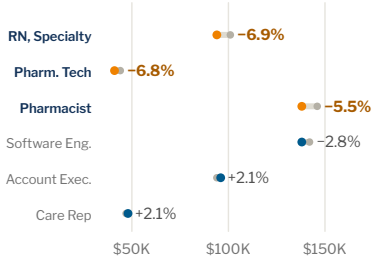



**Recommended**
Grey market, blue McKesson.

## Row order




**Try to avoid**
Alphabetical rows.




**Recommended**
Largest shortfall first, gap labelled.

# Heat table

A table for looking things up, with one column shaded so the pattern shows before anyone reads a number.

Use it for scorecards and quarterly workforce packs: several measures in different units, which readers will copy into their own work. If there is one message about one measure, make the chart instead.

### McKesson spec

- Two shaded columns at most, blue by default
- Horizontal hairlines only; no boxes or zebra stripes
- Twelve rows and seven data columns at most
- No cell under 10 people

1. **Insight headline**
   Says what the table shows.
2. **Subtitle**
   Scope, period, and how totals are built.
3. **Focus rows**
   Bold navy across the whole row.
4. **One shaded column**
   Blue classes, light to dark; text flips to white on dark fills.
5. **Inline bars**
   Size at a glance, from a common edge.
6. **Delta column**
   Signed; only the worse direction in orange.
7. **Units in the headers**
   Bare, right-aligned numbers, same decimals per column.
8. **Weighted total, last**
   Under a navy rule; says what it covers.
9. **Footnote and source**
   The shading key and definitions.

## The two largest frontline job families combine the highest attrition with the lowest engagement (1)

Workforce scorecard by job family, FY26; totals weighted by headcount (2)

| Job family | Headcount (5) | Voluntary attrition (%) (4) | Engagement index | Median days to fill (7) | Change vs. FY25 (days) (6) |
|---|---|---|---|---|---|
| **Pharmacy Tech (3)** | **6,200** | **27.4** | **68** | **52** | **+11** |
| **DC Associate** | **11,800** | **24.1** | **66** | **24** | **−2** |
| Customer Care | 3,100 | 18.9 | 71 | 29 | −2 |
| Specialty Nursing | 1,400 | 14.2 | 74 | 71 | +8 |
| Data & Analytics | 850 | 12.9 | 77 | 58 | +9 |
| Supply Chain | 900 | 11.8 | 75 | 47 | +2 |
| Sales | 2,300 | 10.6 | 76 | 45 | +1 |
| Software Eng. | 2,600 | 9.8 | 74 | 49 | −3 |
| **All 8 job families (8)** | **29,150** | **20.7** | **69** | **38** | **+2** |

Attrition shaded in 5-point classes from under 5% to 25% and above. Orange change = slower to fill than FY25.
Source: Workday HCM, Workday Recruiting and engagement survey (synthetic data for illustration), FY26; U.S. regular employees (9)

RANKING

# Heat table best practice

Formatting should point at the message, not decorate every cell.

**Shade one or two columns.** A color scale on every column is a Christmas tree: nothing stands out.

**No traffic lights.** Red, amber and green fail for color-blind readers and aren't McKesson colors. Use blue classes; orange only when darker means worse.

**Hairlines only.** Boxes and zebra stripes add ink between the numbers. A header rule and light row lines are enough.

**Weight the total.** An average of the rates (19.5) is not the rate for the group (23.3, weighted by headcount).

## How much to shade

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| Pharm. Tech | 27.4 | 68 | 52 |
| DC Associate | 24.1 | 66 | 24 |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |

**Try to avoid**
Every column on its own scale.

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| **Pharm. Tech** | **27.4** | **68** | **52** |
| **DC Associate** | **24.1** | **66** | **24** |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |

**Recommended**
One column; focus rows bold.

## Color scale

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| Pharm. Tech | 27.4 | 68 | 52 |
| DC Associate | 24.1 | 66 | 24 |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |

**Try to avoid**
Red, amber, green.

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| **Pharm. Tech** | **27.4** | **68** | **52** |
| **DC Associate** | **24.1** | **66** | **24** |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |

**Recommended**
Blue classes, light to dark.

## Rules

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| Pharm. Tech | 27.4 | 68 | 52 |
| DC Associate | 24.1 | 66 | 24 |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |

**Try to avoid**
Boxes and zebra stripes.

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| **Pharm. Tech** | **27.4** | **68** | **52** |
| **DC Associate** | **24.1** | **66** | **24** |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |

**Recommended**
Header rule, hairlines.

## Total row

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| **Pharm. Tech** | **27.4** | **68** | **52** |
| **DC Associate** | **24.1** | **66** | **24** |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |
| **Average** | **19.5** | | |

**Try to avoid**
An average of averages.

| | Attrition (%) | Engage-ment | Days to fill |
|---|---|---|---|
| **Pharm. Tech** | **27.4** | **68** | **52** |
| **DC Associate** | **24.1** | **66** | **24** |
| Customer Care | 18.9 | 71 | 29 |
| Spec. Nursing | 14.2 | 74 | 71 |
| Data & Analytics | 12.9 | 77 | 58 |
| **All 5 families** | **23.3** | | |

**Recommended**
Weighted by headcount.

RANKING

# **Build** ranking charts

Native, editable charts and tables in every tool. Tested code and click-by-click steps are on the cookbook pages.

slope-bump.md · dot-plot-dumbbell.md · tables.md

### Before you send it

- [ ] Rank 1 on top; names at both ends
- [ ] One or two highlights, matching the headline
- [ ] Dots, not cut bars, for values far from zero
- [ ] Units in table headers; two shaded columns at most
- [ ] Totals weighted by headcount
- [ ] Nothing ranked or shown under 10 people

| | |
|---|---|
| **PowerPoint and Excel** | **Slope and bump:** Insert › Line with Markers; vertical axis *Values in reverse order*, bounds 0.5 to n + 0.5, labels none; context series grey 1 pt, focus 2.25 pt blue; Series Name labels on the first and last points. **Dumbbell:** XY scatter with custom horizontal error bars (no cap, 3 pt, light sand) as the connector. **Heat table:** clear the table style; bottom borders only; one conditional-formatting rule per blue class on the shaded column; data bars with *Show Bar Only* in a helper column. |
| **Power BI** | Bump: a Line chart on a `RANKX` measure with *Y-axis › Invert range* and series labels on. Dumbbell: the Deneb visual, or a line chart with markers and error bars. Heat table: a Table visual with *Cell elements › Background color › fx › Rules* on one measure. Blank small groups: `IF([Headcount] < 10, BLANK(), [Rank])`. |
| **Python** | matplotlib with `mckviz`; colors from `mv.highlight`, signed gaps from `mv.signed`, cell text from `mv.text_on(fill)`. |
| **HTML** | Plotly `lines+markers+text` with `autorange: "reversed"` for bumps; one connector trace with `None` breaks for dumbbells. Tables as real `<table class="mck-table">` markup with `is-focus`, `is-total` and `.bar`. |
| **R** | `geom_line()` + `geom_point()` with `scale_y_reverse()`; `geom_segment()` under two `geom_point()` layers for dumbbells; `gt` with `data_color(method = "bin")` for heat tables. |

```
colors = dict(zip(ranks, mv.highlight(ranks, focus=risers)))
for name, (r0, r1) in ranks.items():          # rank 1 at the top
    ax.plot([0, 1], [r0, r1], color=colors[name])
    ax.text(-0.05, r0, f"{name}  {r0}", ha="right", va="center")
    ax.text(1.05, r1, f"{r1}  {name}", ha="left", va="center")
ax.set_ylim(len(ranks) + 0.5, 0.5); ax.axis("off")
```

DISTRIBUTION

# **Strip** and box plot

Distribution charts show how a measure spreads across people, teams or sites: what is typical, how wide the range is, and where the tails sit.

Average compa-ratio is 0.97, yet 25% of pharmacy technicians are below 0.90. A strip plot shows every person as a dot; the box marks the middle half.

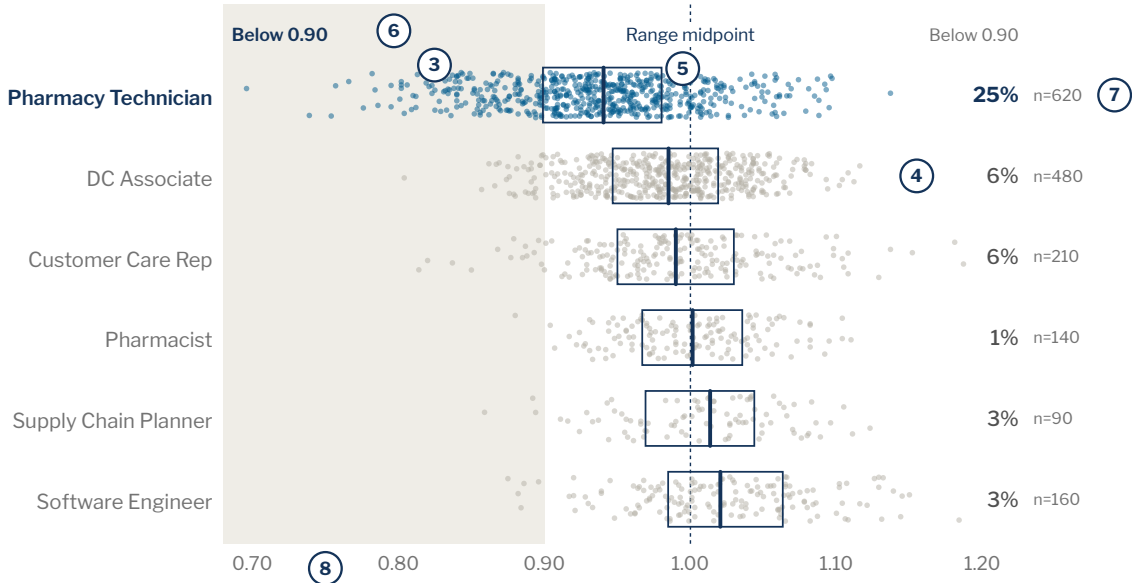
### McKesson spec

- Two to eight groups, each n ≥ 10, n shown
- Grey translucent dots; the focus group blue
- Zone shaded; reference line labelled
- Anonymous dots: no names, IDs or values

1. **Insight headline**
   The share in the tail, and how it compares.
2. **Subtitle**
   Defines the measure, scope and date.
3. **Focus group in blue**
   Each dot one employee; overlaps read as density.
4. **Grey context**
   Every other group, the same grey.
5. **Box on top**
   Middle 50% of people; the thick line is the median.
6. **Zone and reference**
   Shaded below 0.90; the midpoint dashed and named.
7. **Share and n**
   The number the headline uses, at the row end.
8. **Keep the value axis**
   Its ticks are how positions are read; no gridlines.
9. **Footnote and source**
   Explain the dots and the box.

## One in four pharmacy technicians is paid below 0.90 of range midpoint, about 4x the share in any other job family (1)

Compa-ratio (base pay ÷ salary-range midpoint) by job family, U.S. regular employees, Mar 2026 (2)




Each dot is one employee. Box = middle 50% of employees; thick line = median.

Source: Workday HCM (synthetic data for illustration), as of 31 Mar 2026; U.S. regular employees (9)

DISTRIBUTION

# Histogram

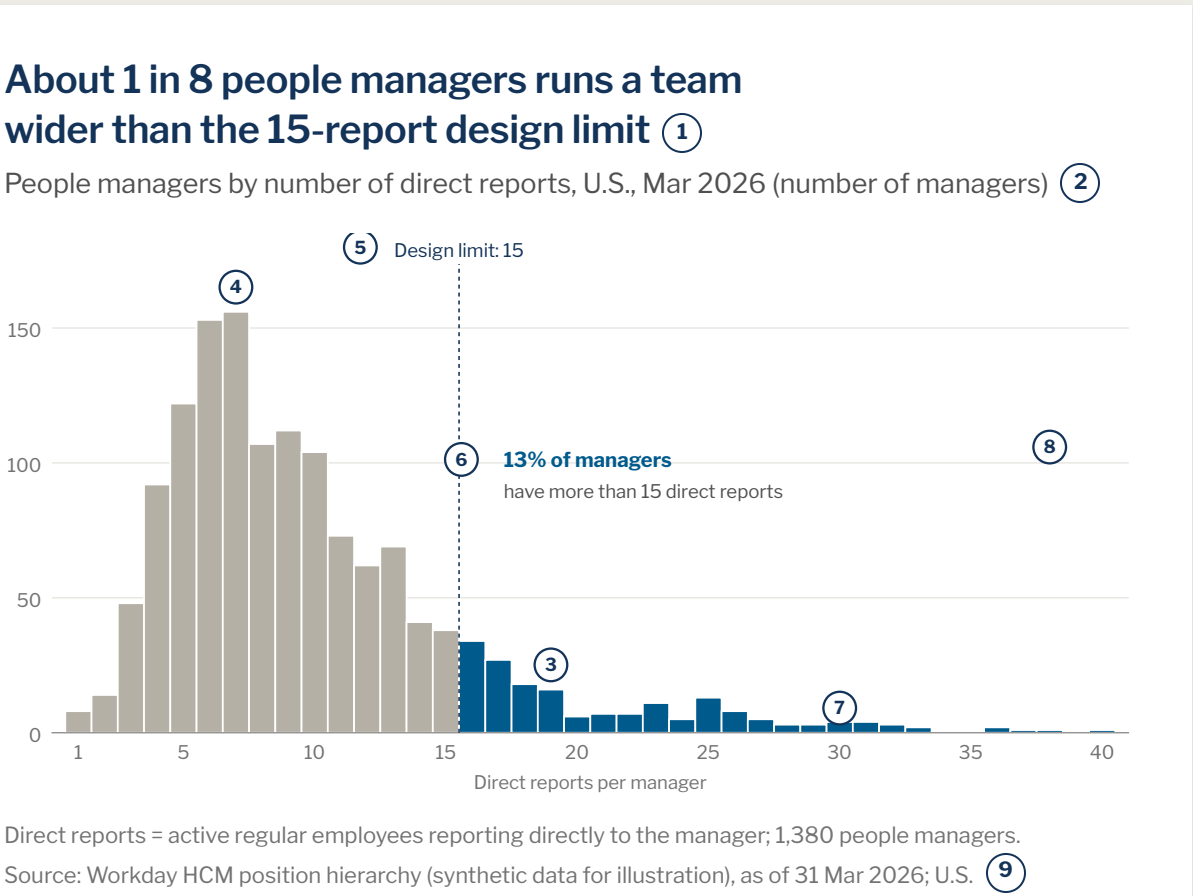
Bars that count how many people fall in each equal-width bin. A histogram shows the shape of one group: skew, a long tail, two peaks.

Use it for one population in detail: span of control across all people managers, days to fill across all requisitions. Here the median manager has 8 direct reports, but the tail runs to 40.

### McKesson spec

- Equal bins that touch, with a thin white seam
- A bin edge exactly on the policy line
- One bin per value for counts; 15–40 bins otherwise
- Bins past the threshold in blue, the rest grey

1. **Insight headline**
   How many people are past the line.
2. **Subtitle**
   Population, date, and what the bars count.
3. **Bins that matter**
   Past the limit, in blue.
4. **Grey context**
   The rest of the distribution.
5. **The limit on a bin edge**
   Dashed navy, named in place.
6. **The share, written out**
   Computed with the same rule as the colors.
7. **One bin per value**
   Whole numbers; bins touch.
8. **Light count gridlines**
   No axis line; the baseline stays.
9. **Footnote and source**
   Define who counts.



DISTRIBUTION

# Distribution variants

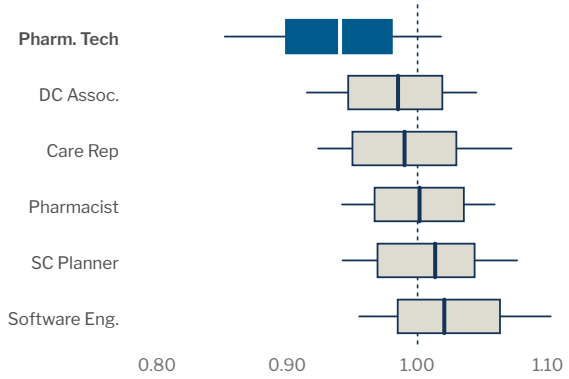
Pick the form by how many groups you compare and how many people are in them.

### When a distribution is the wrong chart

- One value per group (median pay by family): a **bar** or **dot plot**
- How the medians shifted between two dates: a **dumbbell**
- Two measures per person: a **scatter**, or a binned **heatmap** for thousands
- Groups under 10 people: merge them or leave them out

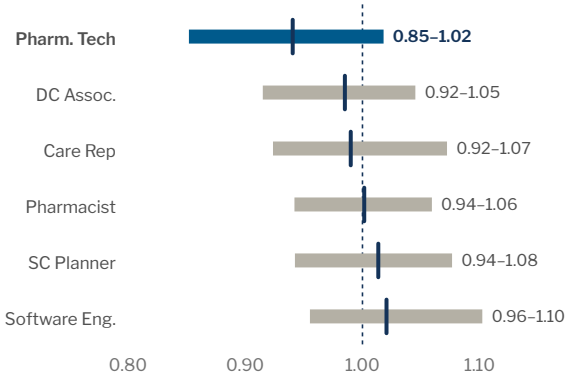
### Box plot

For many groups or tens of thousands of people. Whiskers at the 10th and 90th percentiles, footnoted.



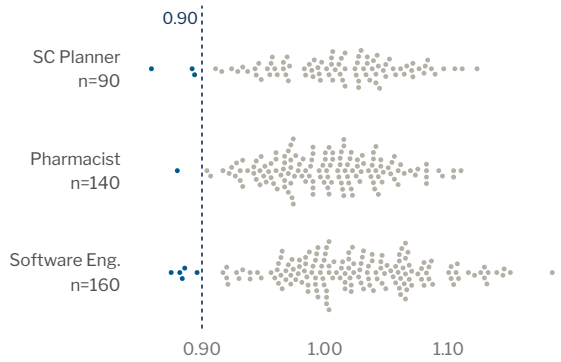

### Range bars

A simpler box for executive decks: the middle 80% as a bar, the median as a tick. Pharmacy technicians start at 0.85.



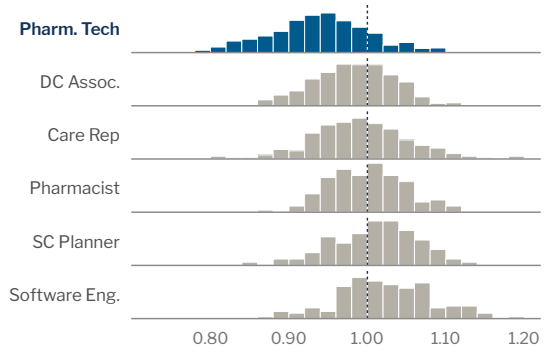

### Beeswarm

Dots that never overlap, so every person is countable. Works up to about 300 per row; above that, jitter.




### Small-multiple histograms

Three to six groups whose shapes differ. Same bins, same scale, shown as a share of each group.



# **Distribution** best practice

The spread is the message; protect it, and protect the people in it.

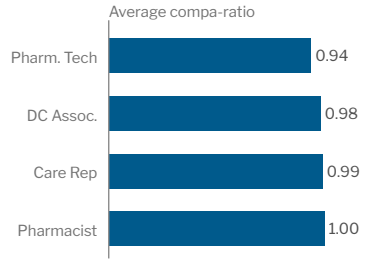
**Show the spread, not just the average.** Group averages look alike; the tails below 0.90 do not.

**A bin edge on the threshold.** Wide bins that straddle 0.90 color the wrong people: 17% instead of 25%.

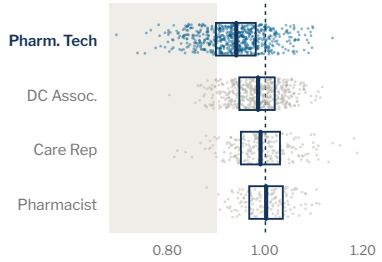
**One scale for every panel.** Rescaled panels make every group look the same shape.

**No group under 10.** A dot in a team of four is a findable person. Merge small groups and show n.

## Spread, not average

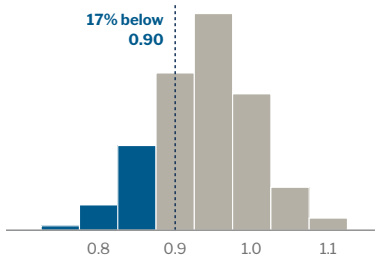



**Try to avoid**
Averages that hide the tail.

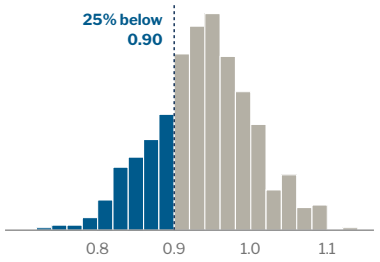



**Recommended**
Every person, with the box.

## Bin edges

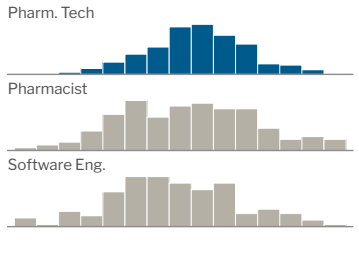



**Try to avoid**
A bin straddling 0.90.

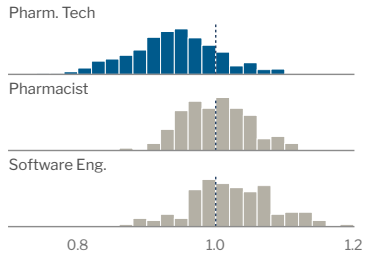



**Recommended**
An edge exactly on 0.90.

## Shared scale

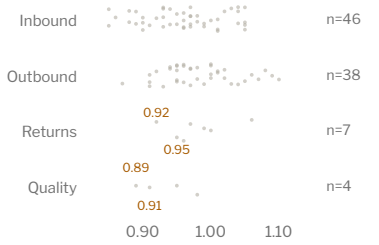



**Try to avoid**
Each panel on its own range.

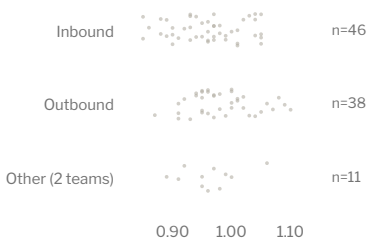



**Recommended**
Same bins, same scale.

## Minimum group size




**Try to avoid**
Teams of 7 and 4, values labelled.




**Recommended**
Small teams merged; n shown.

# **Distribution** best practice, continued

Four more, for strip and box plots in particular.

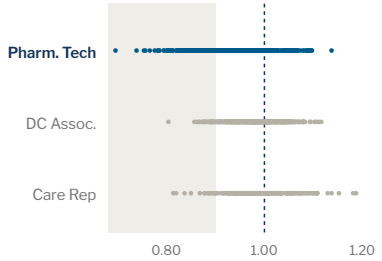
**Jitter and fade.** Opaque dots on one line turn into a bar. Spread them across the row and make them translucent.

**Explain the box.** Few readers know 1.5 × IQR whiskers. Use the 10th–90th percentile and say so.

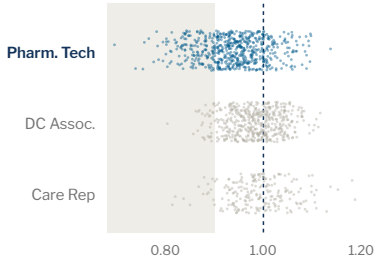
**Don't clip the tails.** A tight axis silently drops people (64 here). Widen it, or cap it and footnote how many.

**Sort by median.** Put the focus group next to the zone, not where the alphabet puts it.

## Overlapping dots

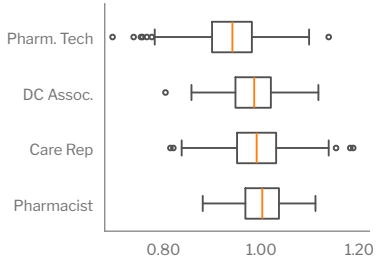



**Try to avoid**
Opaque dots on one line.

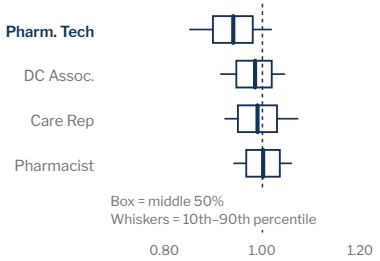



**Recommended**
Jittered and translucent.

## Box and whiskers

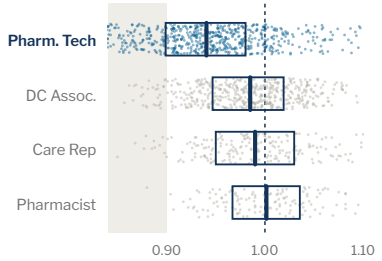



**Try to avoid**
Software defaults, unexplained.

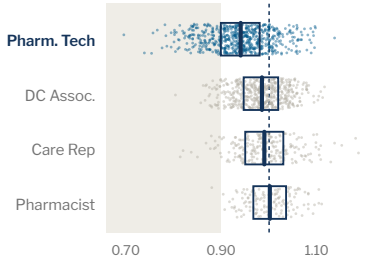



**Recommended**
10th–90th whiskers, explained.

## Axis range

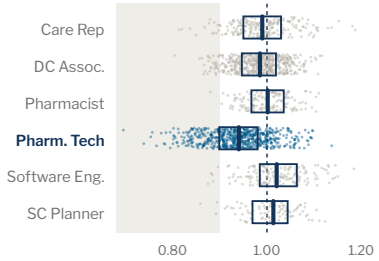



**Try to avoid**
Tails cut off silently.

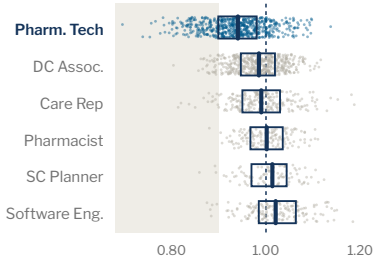



**Recommended**
Room for everyone.

## Group order




**Try to avoid**
Alphabetical.




**Recommended**
By median, focus by the zone.

DISTRIBUTION

# **Build** distribution charts

Set the bins and the jitter yourself, not the software. Tested code and click-by-click steps are on the cookbook page.

distribution.md

### Before you send it

- [ ] Form matches the number of groups
- [ ] Equal bins; an edge on the threshold
- [ ] n for every group; none under 10
- [ ] No individual's value in labels or tooltips
- [ ] One x scale across rows and panels
- [ ] Box footnoted; nothing clipped

| | |
|---|---|
| **PowerPoint and Excel** | **Histogram:** make your own bin column and count with `COUNTIFS`, then a clustered column with Gap Width 0% and white 0.75 pt borders; bins past the threshold in Focus blue. **Box plot:** Insert › Statistic Chart › Box and Whisker, with outlier points and mean markers off. **Strip:** XY scatter with y = group index + `RAND()` jitter, then Paste Values so it stays fixed. |
| **Power BI** | Histogram: *New group › Bin type: Size of bins*, a clustered column chart with inner padding 0, colors by *fx › Rules*, and the threshold as an X-axis constant line. No native box plot: use the certified Box and Whisker visual or a scatter with a jitter column. Turn tooltips off for individual points. |
| **Python** | matplotlib with `mckviz`: fixed-seed jitter, translucent dots, the box drawn on top. |
| **HTML** | Plotly: pre-bin with `np.histogram` and draw `go.Bar` so you control the edges; `go.Box` with `boxpoints: "all"` for strips, and `hoveron: "boxes"` so no one can hover a single person's pay. |
| **R** | `geom_jitter()` under `geom_boxplot(fill = NA, outlier.shape = NA)`; `geom_histogram(breaks = ...)` with explicit edges; `facet_wrap()` keeps one scale. |

```
rng = np.random.default_rng(1)                # fixed jitter: reproducible
for y, fam in enumerate(order):               # groups with n >= 10 only
    x = df.compa_ratio[df.job_family == fam]
    ax.scatter(x, y + rng.uniform(-0.28, 0.28, len(x)), s=7, alpha=0.45,
               color=C.focus if fam == focus else C.context, lw=0)
ax.axvspan(0.68, 0.90, color=C.surface_subtle, zorder=0)
```